Note:

The RK806S-5 LDO power distribution of the reference schematic is only suitable for the interface used in the reference schematic.
If other interface functions need to be added to the reference schematic, the RK806S-5 LDO distribution needs to be re evaluated, otherwise the added functions may exceed the maximum current provided by the LDO

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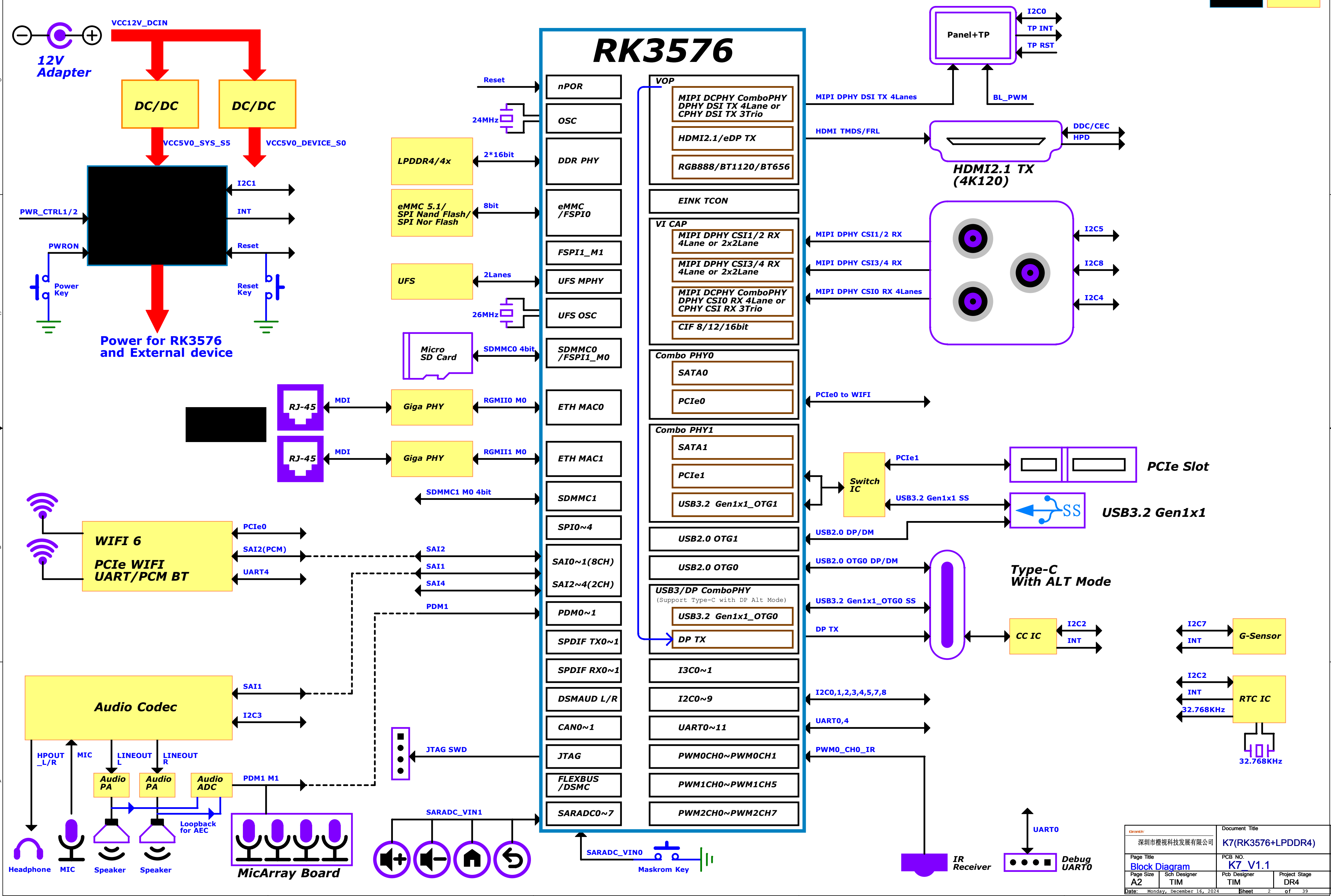
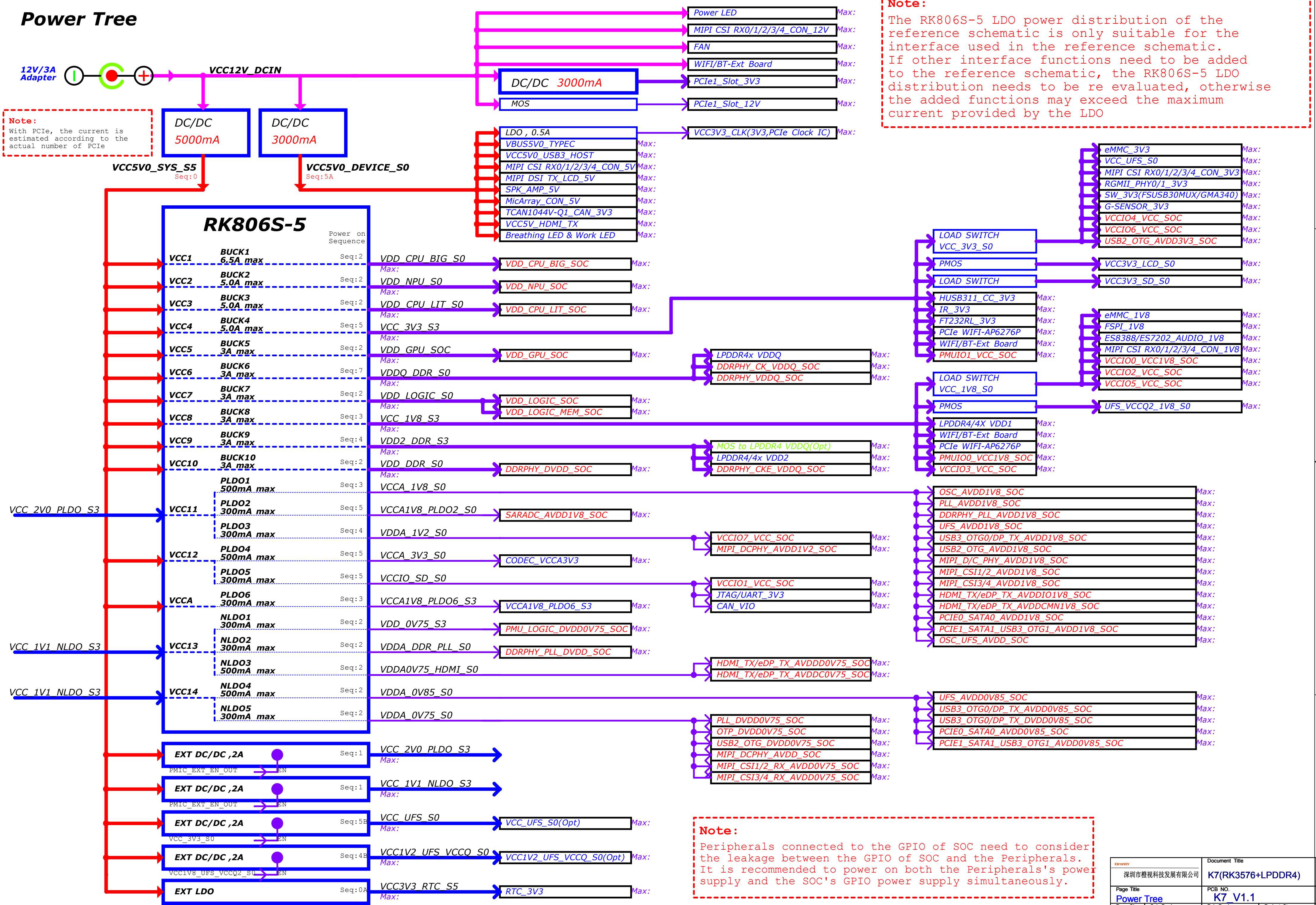
RK3576 Ref Block Diagram(Typical Application Case)



Power Tree



Note:
With PCIe, the current is estimated according to the actual number of PCIe

Note:
The RK806S-5 LDO power distribution of the reference schematic is only suitable for the interface used in the reference schematic.
If other interface functions need to be added to the reference schematic, the RK806S-5 LDO distribution needs to be re evaluated, otherwise the added functions may exceed the maximum current provided by the LDO

Note:
Peripherals connected to the GPIO of SOC need to consider the leakage between the GPIO of SOC and the Peripherals.
It is recommended to power on both the Peripherals's power supply and the SOC's GPIO power supply simultaneously.

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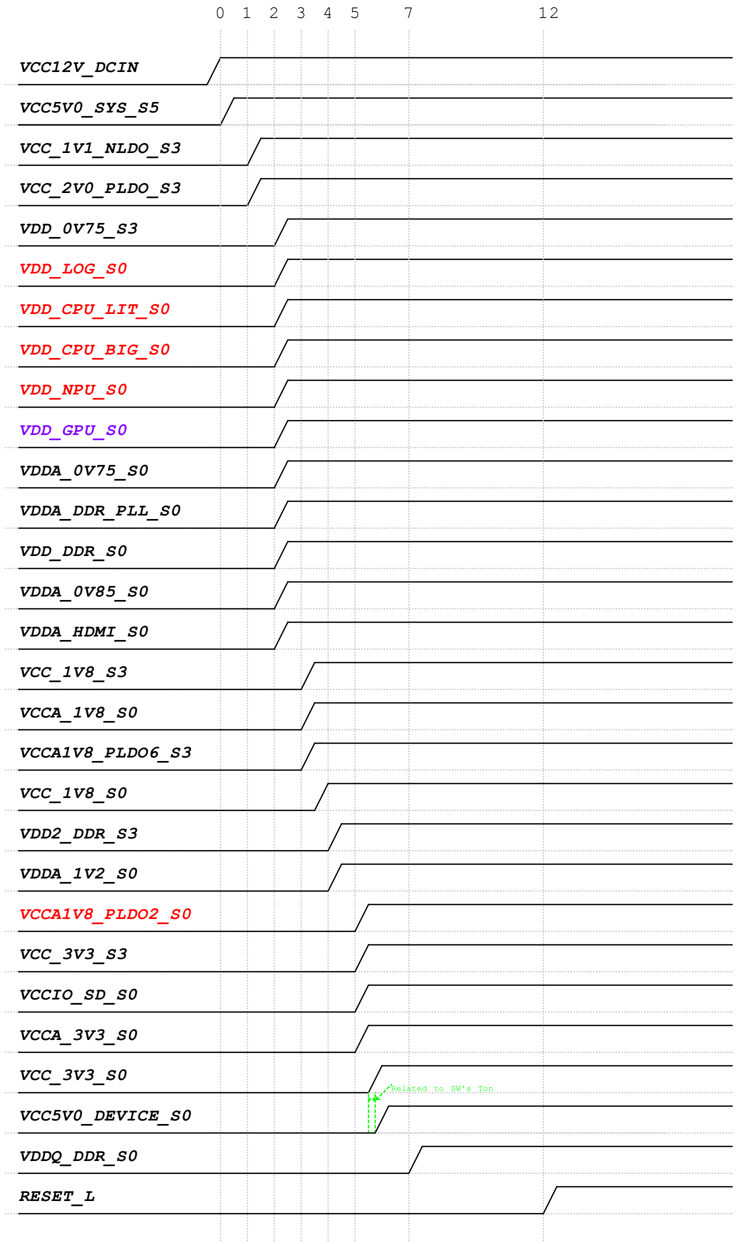
Power Sequence



Power description

Power Supply	PMIC Channel	Supply Limit	Power Name	Time Slot	Default Voltage	Default ON/OFF	Work Voltage	Peak Current	Sleep Current
VCC5V0_SYS_S5	RK806_BUCK1	6.5A	VDD_CPU_BIG_S0	Slot:2	0.85V	ON	DVFS	TBD	TBD
VCC5V0_SYS_S5	RK806_BUCK2	5A	VDD_NPU_S0	Slot:2	0.75V	ON	DVFS	TBD	TBD
VCC5V0_SYS_S5	RK806_BUCK3	5A	VDD_CPU_LIT_S0	Slot:2	0.85V	ON	DVFS	TBD	TBD
VCC5V0_SYS_S5	RK806_BUCK4	5A	VCC_3V3_S3	Slot:5	3.3V	ON	3.3V	TBD	TBD
VCC5V0_SYS_S5	RK806_BUCK5	3A	VDD_GPU_S0	Slot:2	ADJ FB=0.5V	ON	DVFS	TBD	TBD
VCC5V0_SYS_S5	RK806_BUCK6	3A	VDDQ_DDR_S0	Slot:7	ADJ FB=0.5V	ON	0.61V-LP4/4x 0.51V-LP5	TBD	TBD
VCC5V0_SYS_S5	RK806_BUCK7	3A	VDD_LOGIC_S0 VDD_LOGIC_MEM_S0	Slot:2	0.75V	ON	0.75V	TBD	TBD
VCC5V0_SYS_S5	RK806_BUCK8	3A	VCC_1V8_S3	Slot:3	1.8V	ON	1.8V	TBD	TBD
VCC5V0_SYS_S5	RK806_BUCK9	3A	VDD2_DDR_S3	Slot:4	ADJ FB=0.5V	ON	1.1V-LP4/4x 1.05V-LP5	TBD	TBD
VCC5V0_SYS_S5	RK806_BUCK10	3A	VDD_DDR_S0	Slot:2	0.85V	ON	0.85V DVFS	TBD	TBD
VCC_2V0_PLDO	RK806_PLDO1	0.5A	VCCA_1V8_S0	Slot:3	1.8V	ON	1.8V	TBD	TBD
	RK806_PLDO2	0.3A	VCCA1V8_PLDO2_S0	Slot:5	1.8V	ON	1.8V	TBD	TBD
	RK806_PLDO3	0.3A	VDDA_1V2_S0	Slot:4	1.2V	ON	1.2V	TBD	TBD
VCC5V0_SYS_S5	RK806_PLDO4	0.5A	VCCA_3V3_S0	Slot:5	3.0V	ON	3.3V	TBD	TBD
	RK806_PLDO5	0.3A	VCCIO_SD_S0	Slot:5	3.3V	ON	3.3V	TBD	TBD
VCC5V0_SYS_S5	RK806_PLDO6	0.3A	VCCA1V8_PLDO6_S3	Slot:3	1.8V	ON	1.8V	TBD	TBD
VCC_1V1_NLDO	RK806_NLDO1	0.3A	VDD_0V75_S3	Slot:2	0.75V	ON	0.75V	TBD	TBD
	RK806_NLDO2	0.3A	VDDA_DDR_PLL_S0	Slot:2	0.85V	ON	0.85V DVFS	TBD	TBD
	RK806_NLDO3	0.5A	VDDA0V75_HDMI_S0	Slot:2	0.75V	ON	0.75V	TBD	TBD
VCC_1V1_NLDO	RK806_NLDO4	0.5A	VDDA_0V85_S0	Slot:2	0.85V	ON	0.85V	TBD	TBD
	RK806_NLDO5	0.3A	VDDA_0V75_S0	Slot:2	0.75V	ON	0.75V	TBD	TBD
	RK806_RESETn								
VCC5V0_SYS_S5	EXT BUCK	2A	VCC_2V0_PLDO_S3	Slot:1	2.1V	ON	2.0V	TBD	TBD
VCC5V0_SYS_S5	EXT BUCK	2A	VCC_1V1_NLDO_S3	Slot:1	1.1V	ON	1.1V	TBD	TBD
VCC12V_DCIN	EXT BUCK	5A	VCC5V0_SYS_S5	Slot:0	5.0V	ON	5.0V	TBD	TBD
VCC12V_DCIN	EXT BUCK	3A	VCC5V0_DEVICE_S0	Slot:5A	5.2V	ON	5.2V	TBD	TBD
VCC_3V3_S3	SWITCH	2A	VCC_3V3_S0	Slot:5A	3.3V	ON	3.3V	TBD	TBD
VCC_1V8_S3	SWITCH	2A	VCC_1V8_S0	Slot:3A	1.8V	ON	1.8V	TBD	TBD

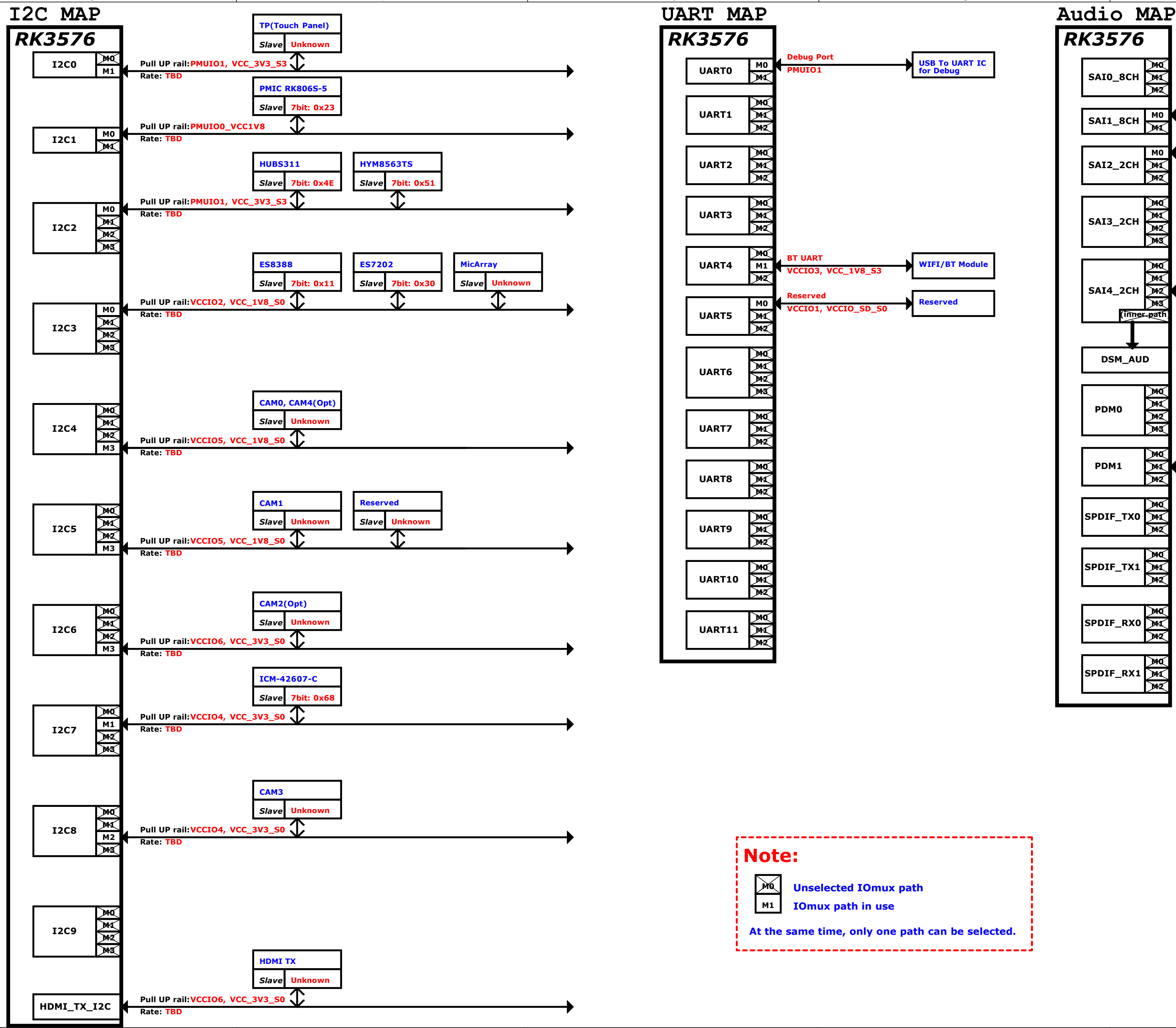
Note:
The power suffix S0, S3 or S5 means:
S5: Keep power on during power down
S3: Keep power on during sleeping
S0: Power off during sleeping

Note:
Peripherals connected to the GPIO of SOC need to consider the leakage between the GPIO of SOC and the Peripherals.
It is recommended to power on both the Peripherals's power supply and the SOC's GPIO power supply simultaneously.

IO Power Domain Map

IO Domain	Pin Num	Support IO Voltage	Supply Power Pin Name	Power Source	Operating Voltage
PMUIO0	Pin 2K11	1.8V Only	PMUIO0_VCC1V8	VCC_1V8	1.8V
PMUIO1	Pin 1U20	1.8V or 3.3V	PMUIO1_VCC	VCC_1V8 VCC_3V3	3.3V
VCCIO0	Pin 1J20	1.8V Only	VCCIO0_VCC1V8	VCC_1V8	1.8V
VCCIO1	Pin 2A8	1.8V or 3.3V	VCCIO1_VCC	VCC_1V8 VCC_3V3	1.8V/3.3V
VCCIO2	Pin 2A2	1.8V or 3.3V	VCCIO2_VCC	VCC_1V8 VCC_3V3	1.8V
VCCIO3	Pin 2B10	1.8V or 3.3V	VCCIO3_VCC	VCC_1V8 VCC_3V3	1.8V
VCCIO4	Pin 2A7	1.8V or 3.3V	VCCIO4_VCC	VCC_1V8 VCC_3V3	3.3V
VCCIO5	Pin 2A4/2A5	1.8V or 3.3V	VCCIO5_VCC	VCC_1V8 VCC_3V3	1.8V
VCCIO6	Pin 2N3	1.8V or 3.3V	VCCIO6_VCC	VCC_1V8 VCC_3V3	3.3V
VCCIO7	Pin 2M3	1.2V or 1.8V	VCCIO7_VCC	VCC_1V2 VCC_1V8	1.2V

IO Type	Operating Voltage
1.8V Only	VCCIO*_VCC1V8=1.8V
1.2V or 1.8V	VCCIO*_VCC=1.2V or 1.8V
1.8V or 3.3V	VCCIO*_VCC=1.8V or 3.3V



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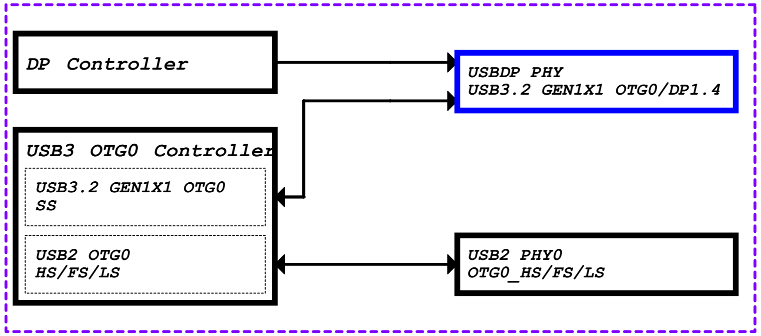
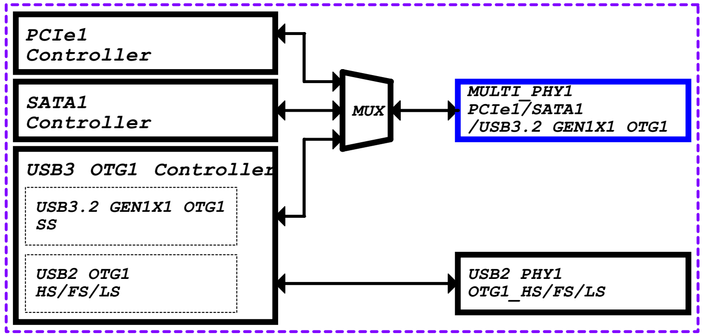
MULTI_PHY Path Map

USBDP PHY--USB3.2 GEN1X1 OTG0/DP1.4



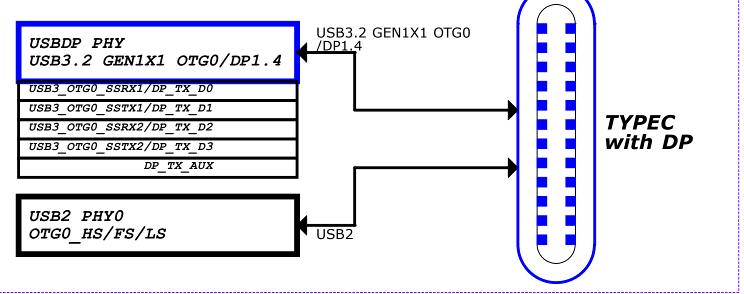
PCIe Combo PHY1--
PCIe1/SATA1/USB3.2 GEN1X1 OTG1



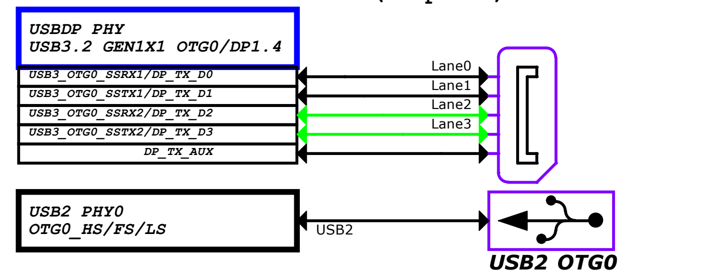
Note:
USB2 PHY1 can only be used when
PCIE1/SATA1 is not in use!!!

USB OTG0/DP Application

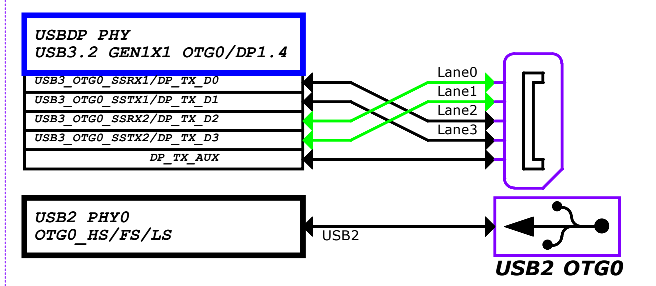
CASE0: TYPEC with DP



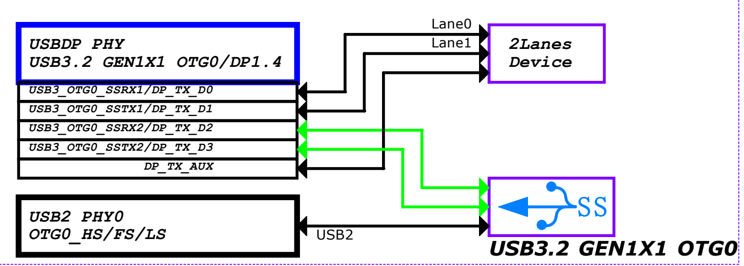
CASE1: USB2 OTG0 + DP 4Lane(Swap OFF)



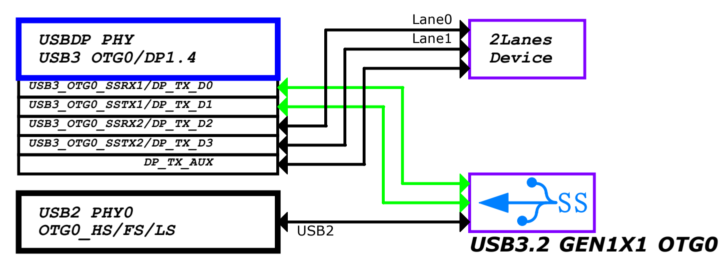
CASE2: USB2 OTG0 + DP 4Lane(Swap ON)



CASE3: USB3.2 GEN1X1 OTG0 + DP 2Lane(Swap OFF)



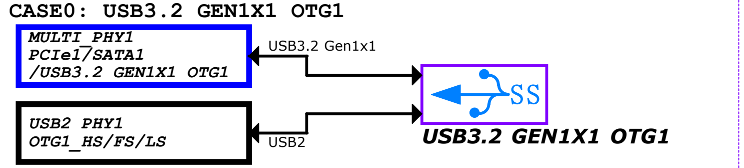
CASE4: USB3.2 GEN1X1 OTG0 + DP 2Lane(Swap ON)



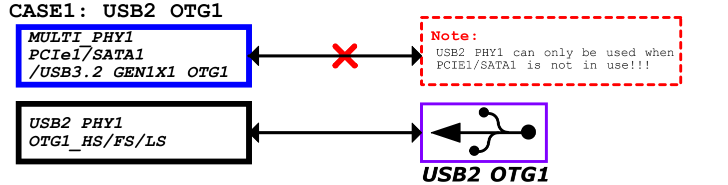
Note:
DP Lane swap enable
0:Lane0/1/2/3 TxData mapping to Lane0/1/2/3_TXDP/N
1:Lane0/1/2/3 TxData mapping to Lane2/3/0/1_TXDP/N

USB OTG1 Application

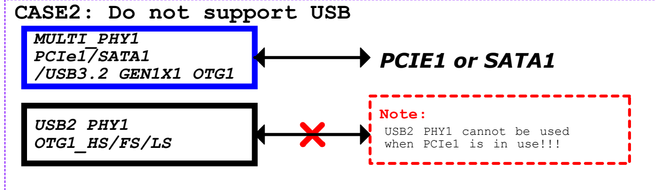
CASE0: USB3.2 GEN1X1 OTG1



CASE1: USB2 OTG1



CASE2: Do not support USB



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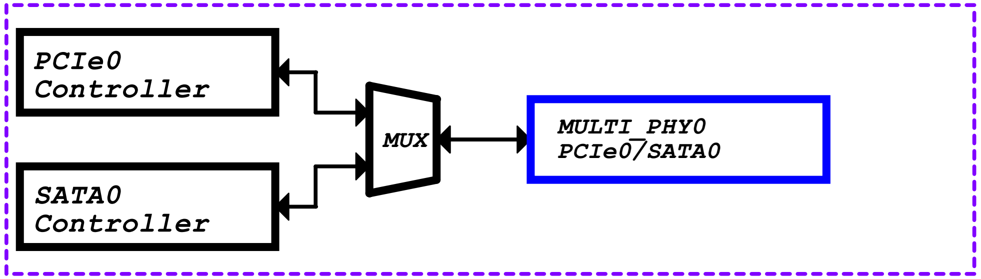
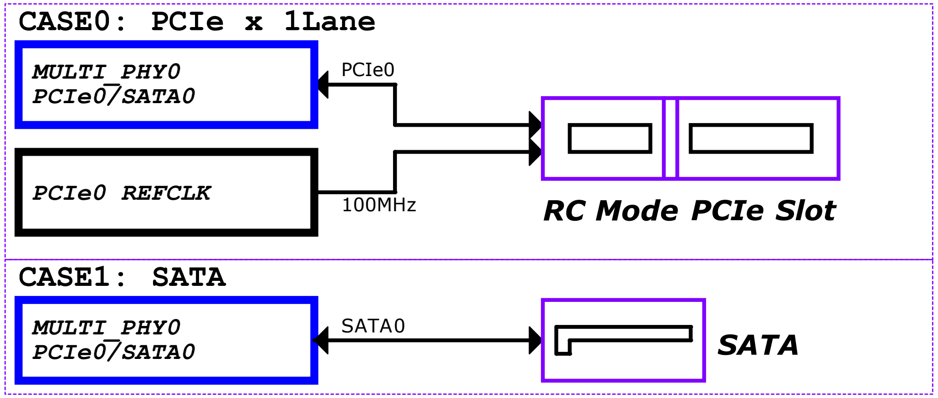
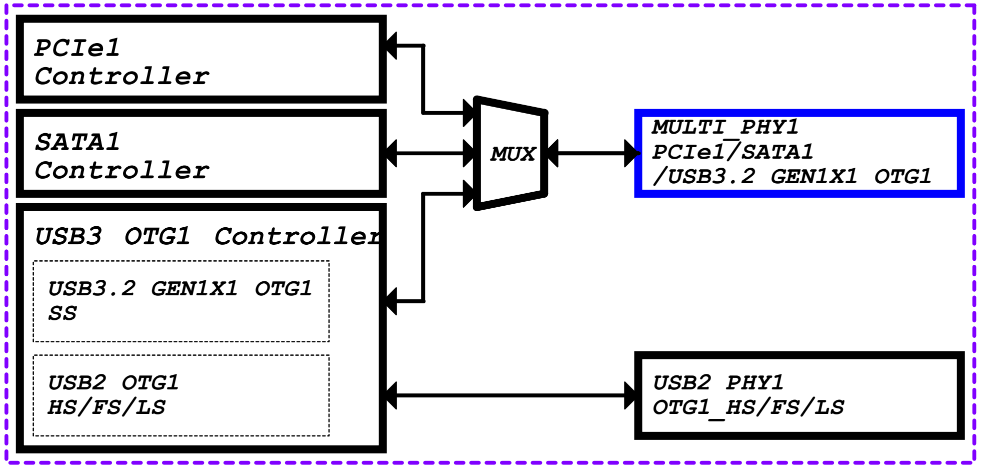
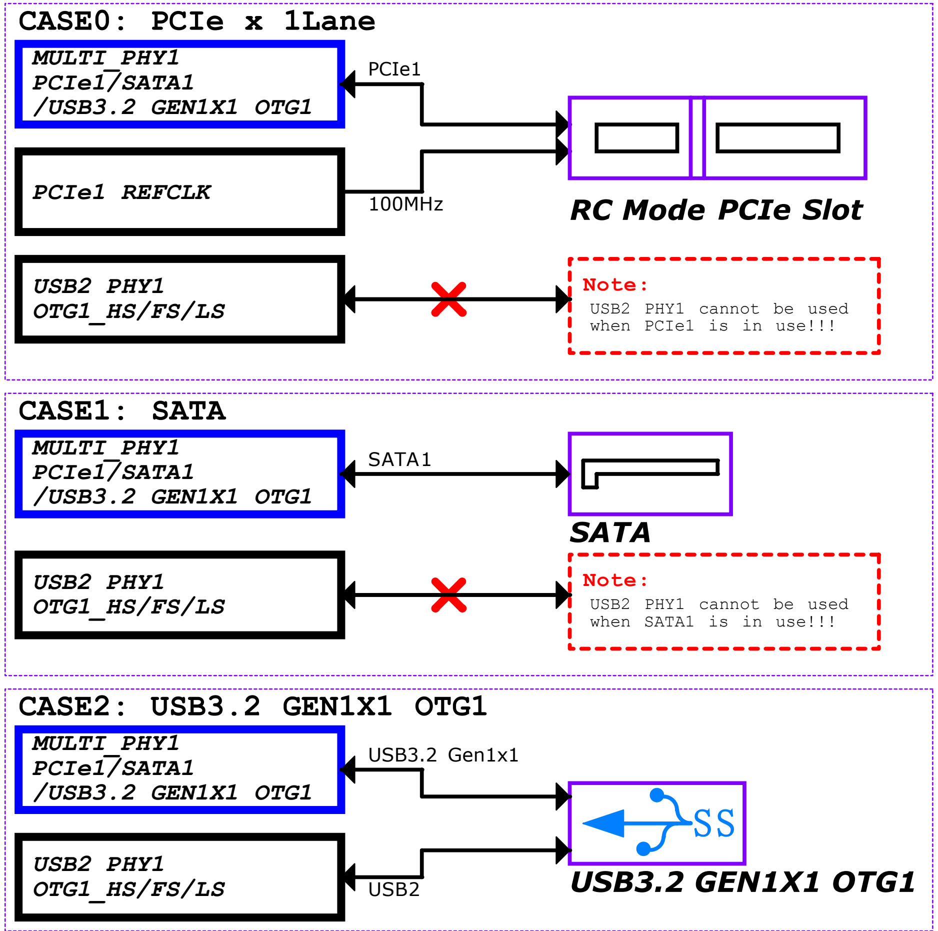
PCIe Combo PHY0--PCIe0/SATA0





PCIe Combo PHY1--PCIe1/SATA1/USB3.2 GEN1X1 OTG1





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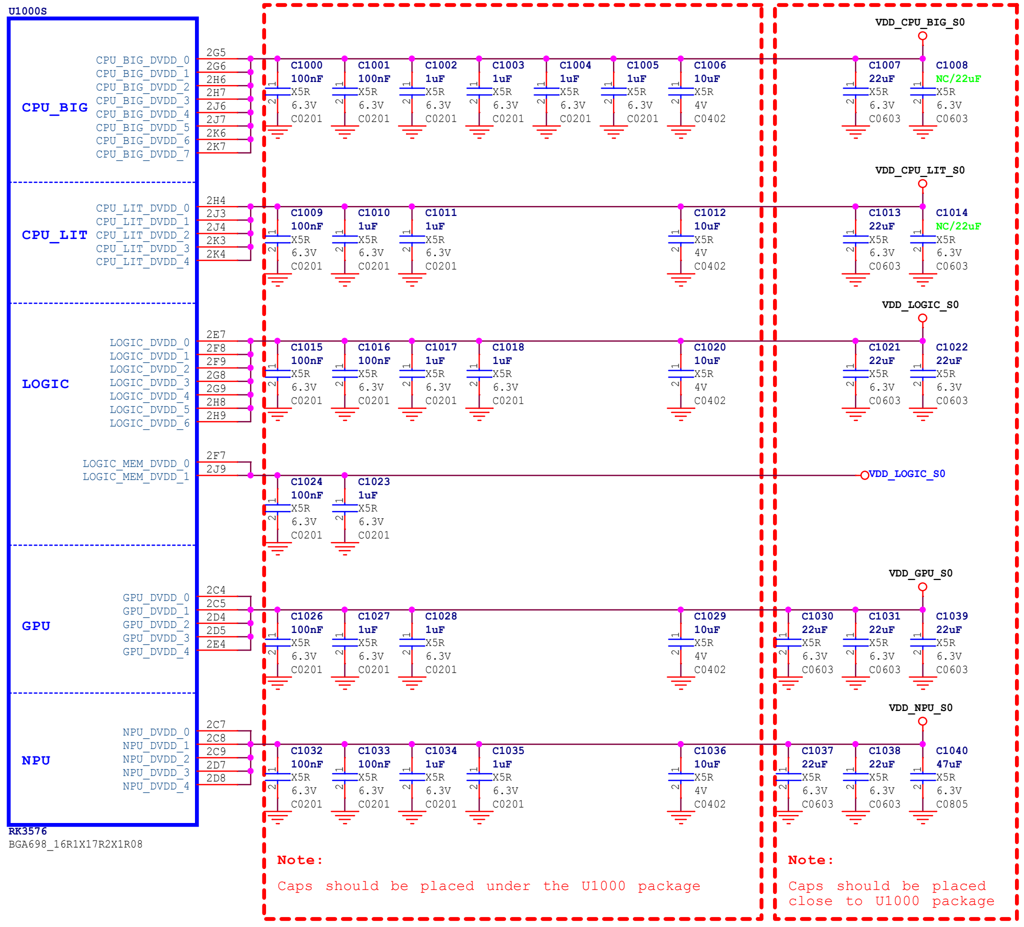
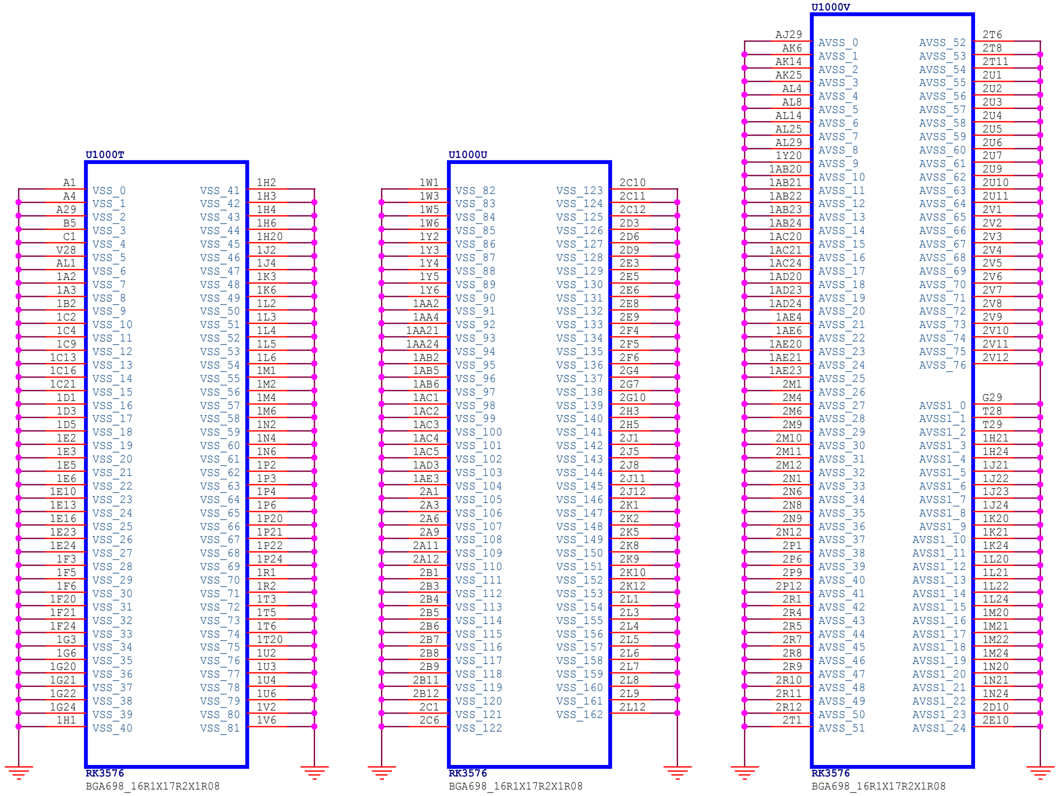
RK3576_S (Power)



RK3576_T/U/V (GND)



Orenth 深圳市橙视科技发展有限公司	Document Title: K7(RK3576+LPDDR4)
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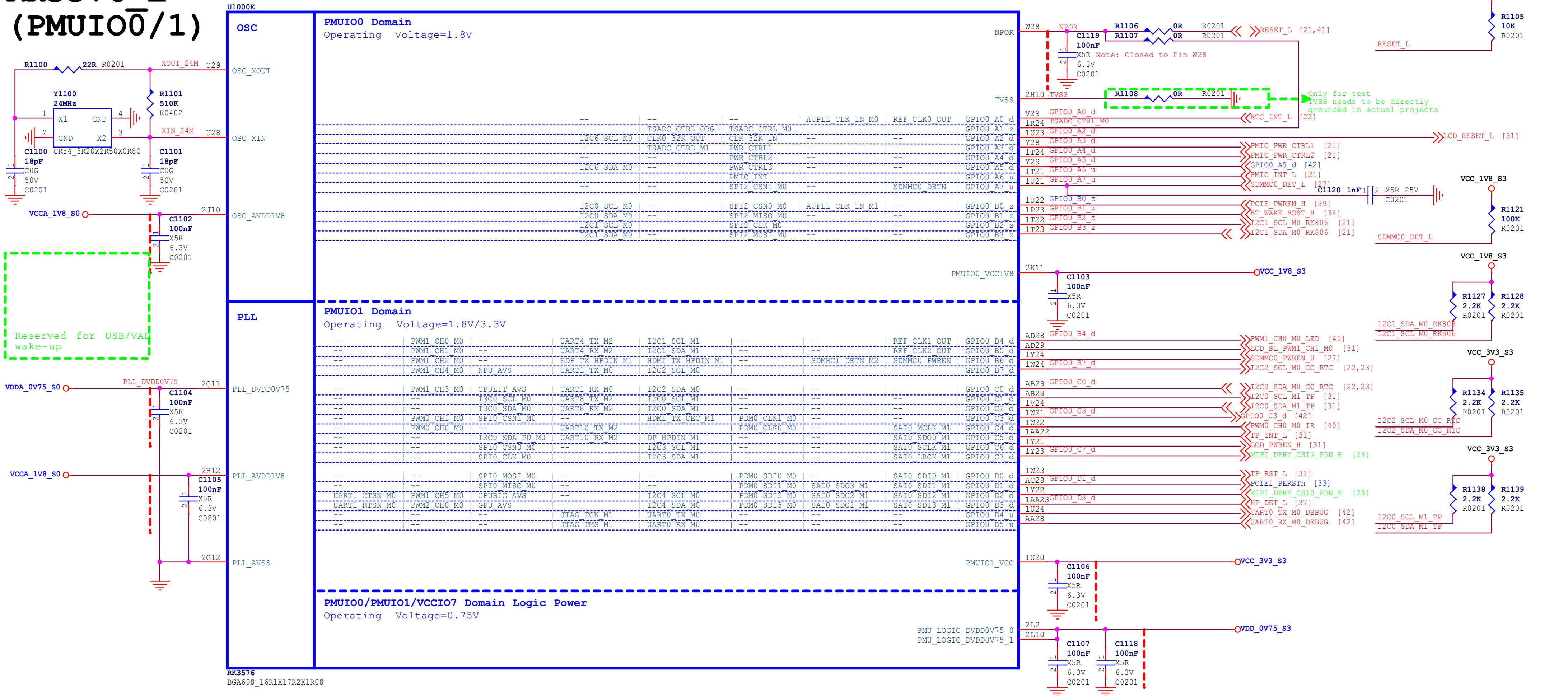
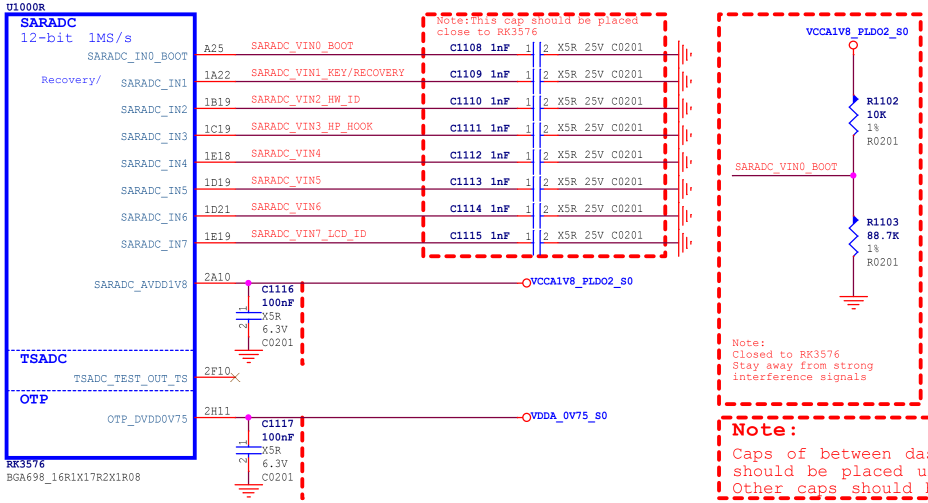
RK3576_E (PMUIO0/1)



RK3576_R (SARADC)

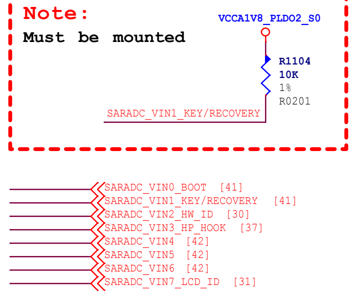


BOOT MODE CONFIG

Config Table for SARADC_VIN0_BOOT

Item	Rup	Rdown	ADC Value	Boot Mode
Config1	NC	10K	0	USB (Maskrom mode)
Config2	10K	1.13K	416	FSPI0->USB
Config3	10K	2.49K	816	FSPI1_M0->EMMC->USB
Config4	10K	4.3K	1231	FSPI1_M1->EMMC->USB
Config5	10K	6.8K	1658	FSPI0->UFS->USB
Config6	10K	10K	2048	FSPI1_M0->UFS->USB
Config7	10K	14.7K	2437	UFS->USB
Config8	10K	23.2K	2862	UFS->SDMMC0->USB
Config9	10K	40.2K	3279	RFU
Config10	10K	88.7K	3680	EMMC->SDMMC0->USB
Config11	10K	NC	4095	EMMC->USB

Note:
Must be mounted



Note:
Caps of between dashed red lines and U1000 should be placed under the U1000 package.
Other caps should be placed close to the U1000 package.

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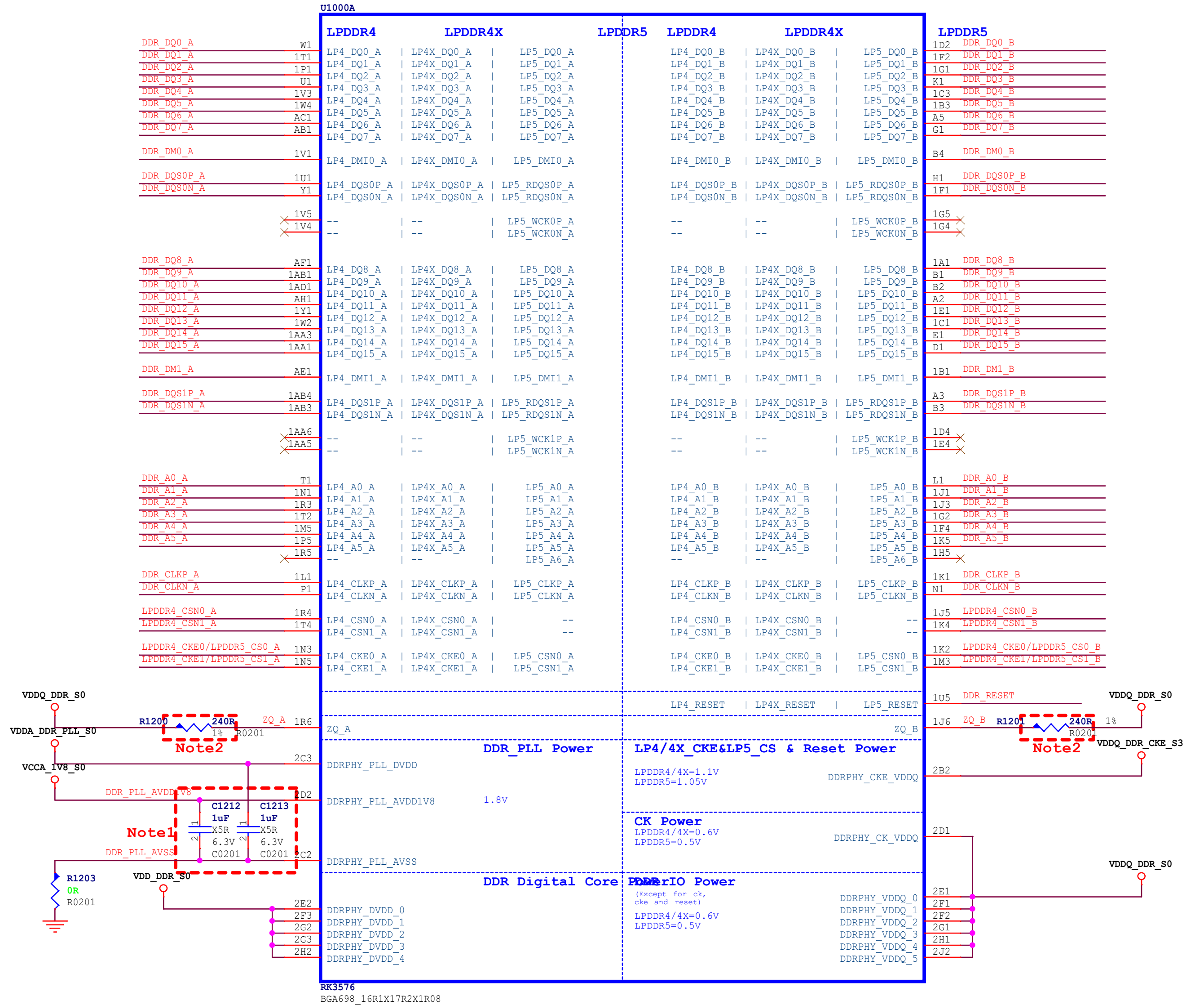
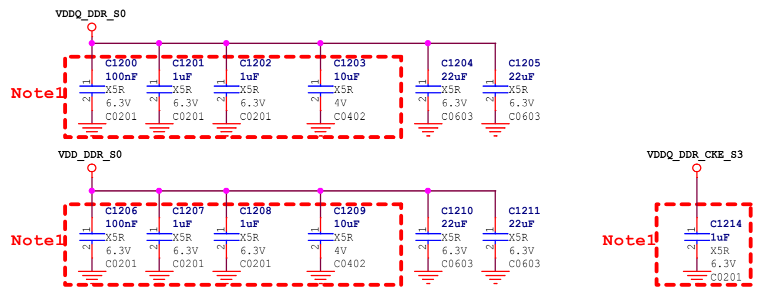
RK3576_A(DDRPHY)



DDR FILTER



Note:

(1) Power Sequence: VDD-VDDQ_CKE-VDDQ
(2) Hold power of DDRPHY_CKE_VDDQ during retention times.

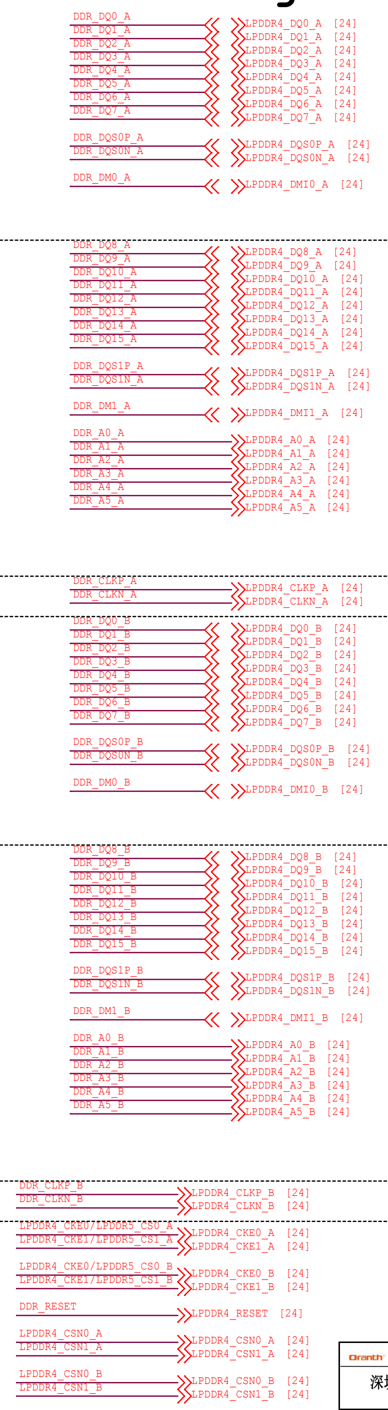
Note1:

Caps in the red line dotted box should be placed under the U1000 package

Note2:

Resistors in the red line dotted box should be placed under the U1000 package

LPDDR4 Signal



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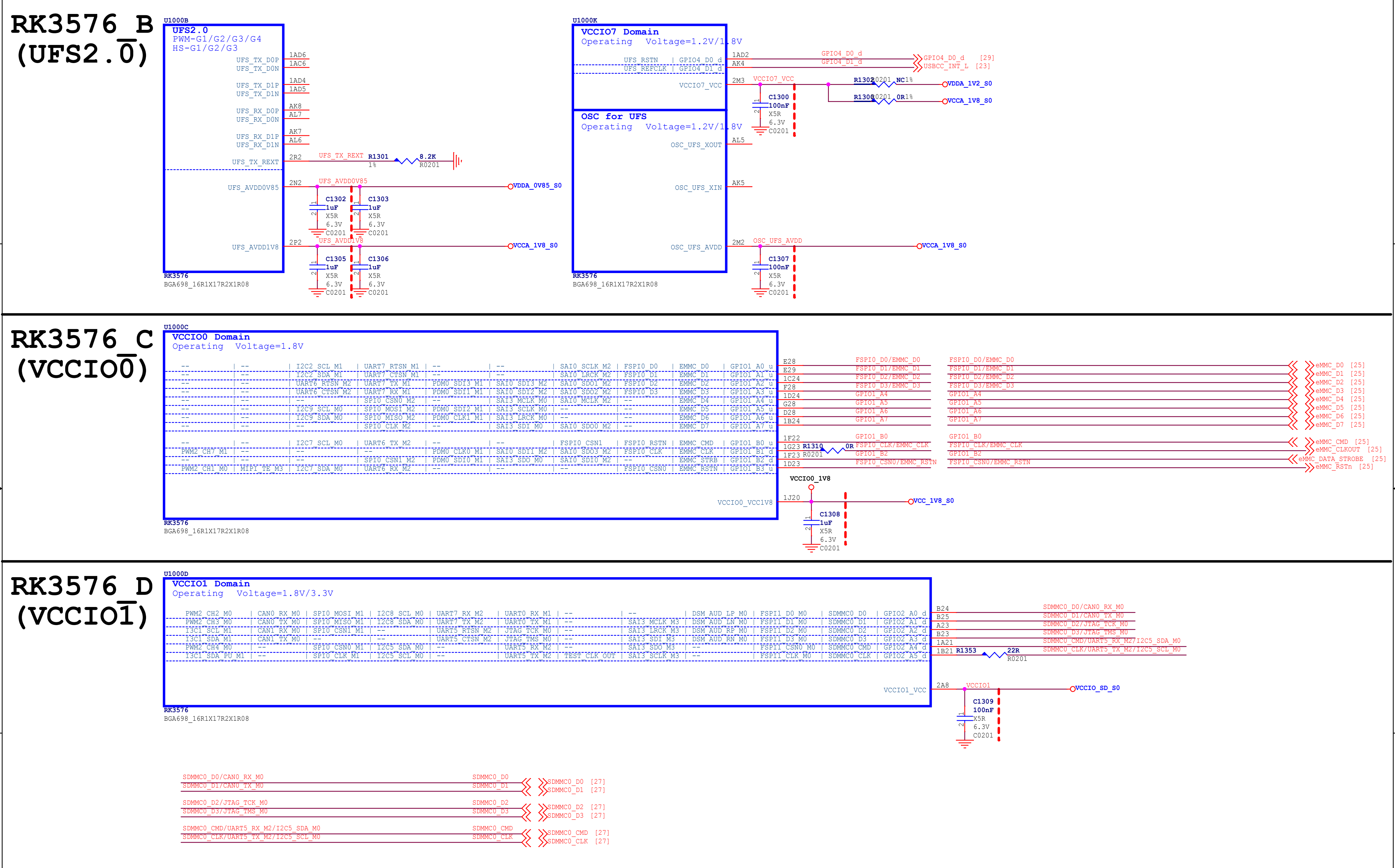

Orenth 深圳市橙视科技发展有限公司	Document Title: K7(RK3576+LPDDR4)
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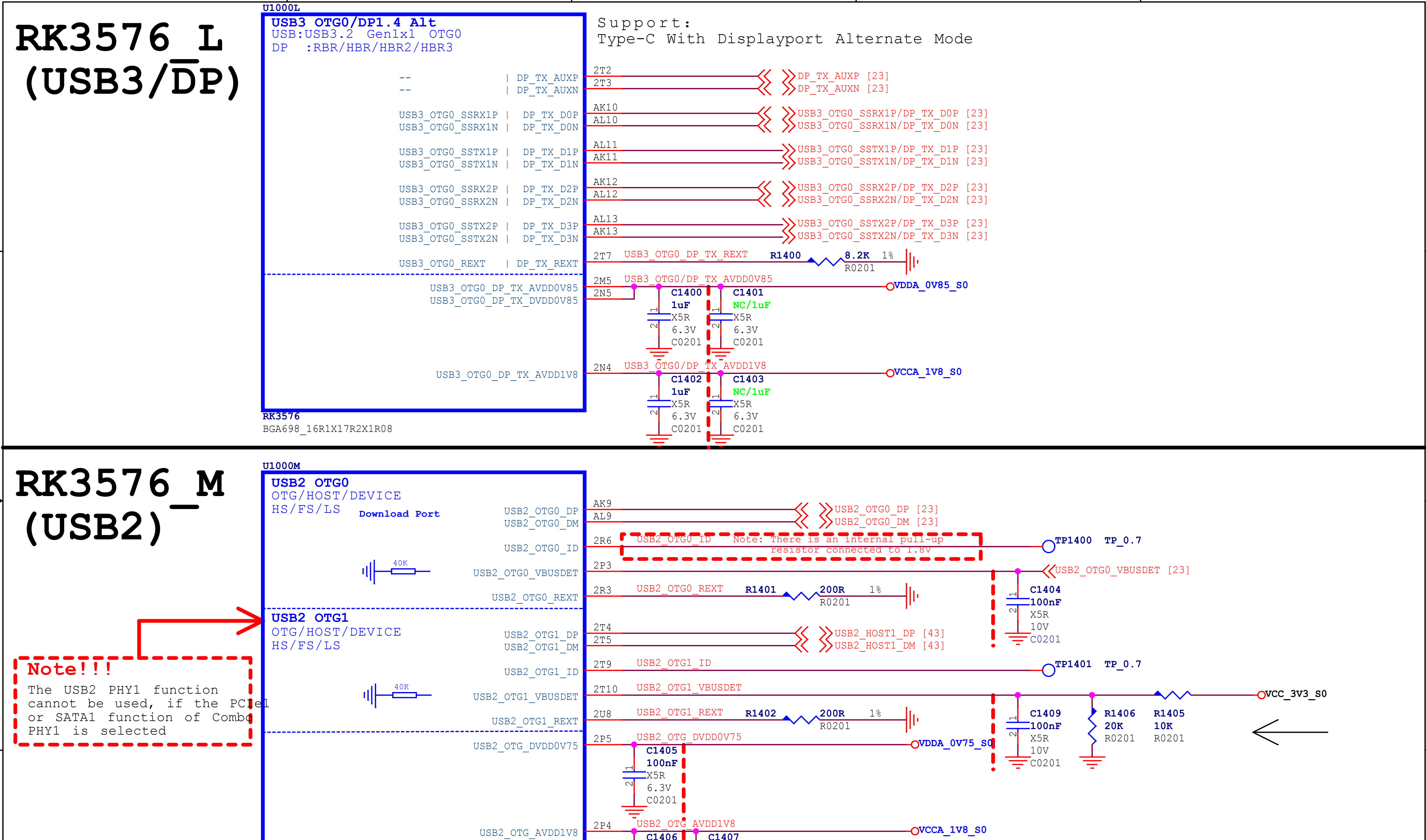

Oranth 深圳市橙视科技发展有限公司	Document Title: K7(RK3576+LPDDR4)		
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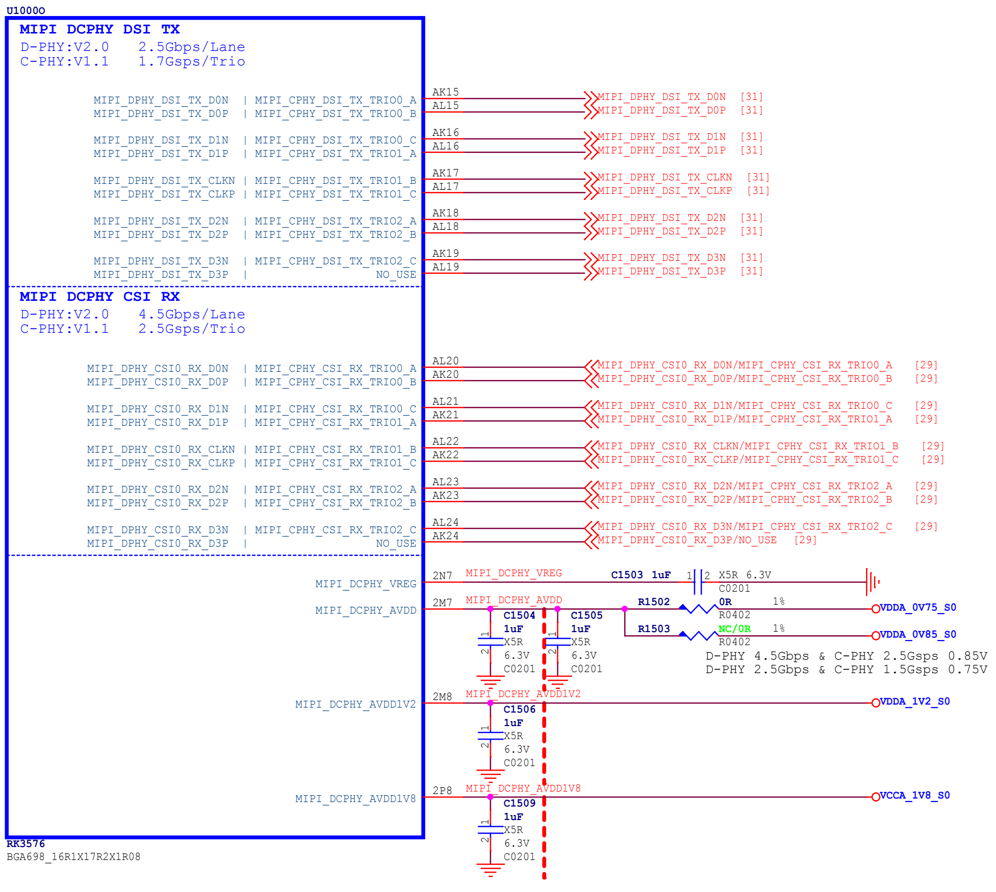
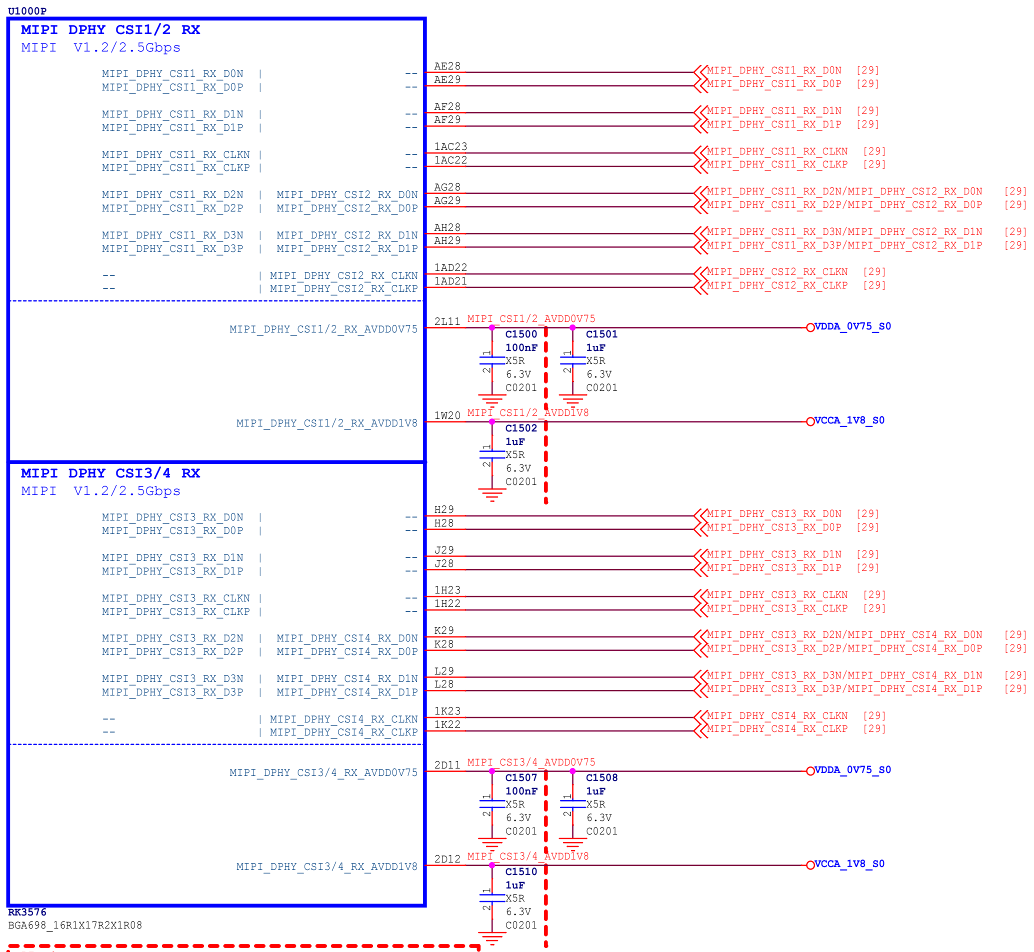
RK3576_O(MIPI DCPHY)



Note:
Caps of between dashed red lines and U1000 should be placed under the U1000 package. Other caps should be placed close to the U1000 package

RK3576_P(MIPI DPHY CSI RX)



Note:
Caps of between dashed red lines and U1000 should be placed under the U1000 package. Other caps should be placed close to the U1000 package

深圳市橙视科技发展有限公司	Document Title: K7(RK3576+LPDDR4)
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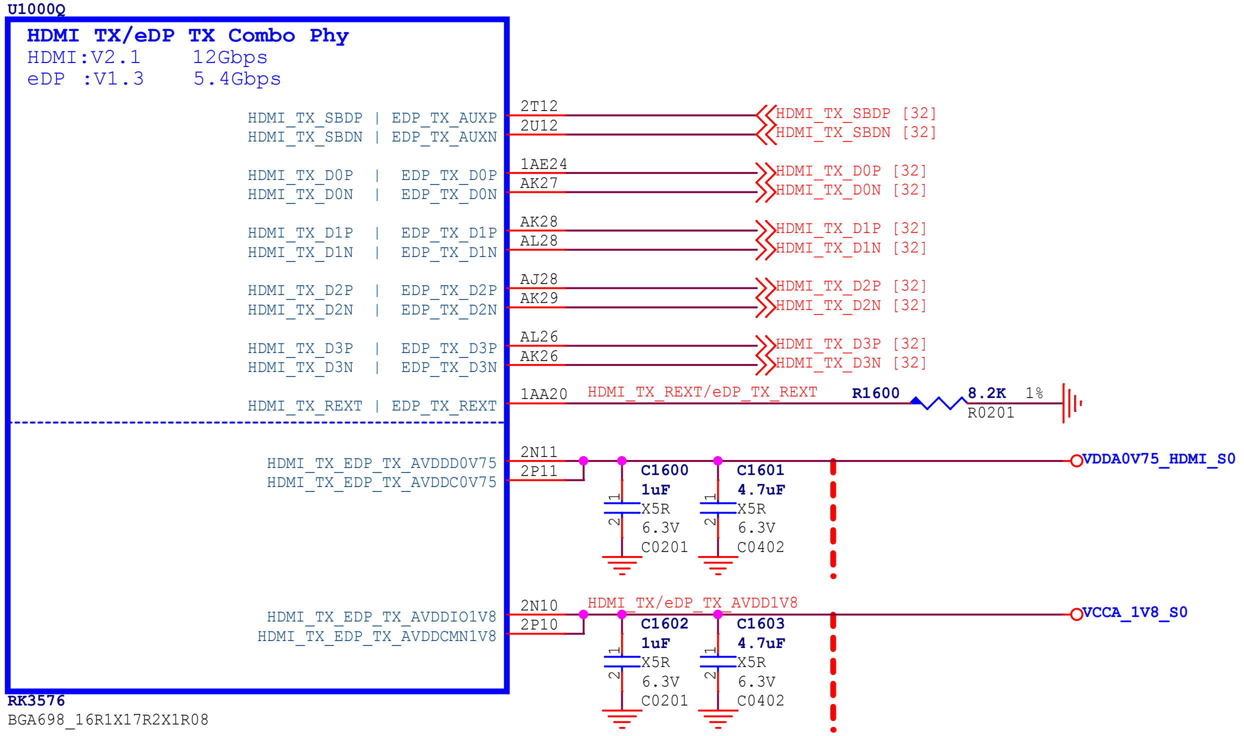
RK3576_Q(HDMI/eDP)

Note:
HDMI 2.1 supports up to 4Kx2K@120Hz



Note:
Caps of between dashed red lines and U1000 should be placed under the U1000 package. Other caps should be placed close to the U1000 package

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Page Title: RK3576-HDMI/eDP	PCB NO.: K7_V1.1
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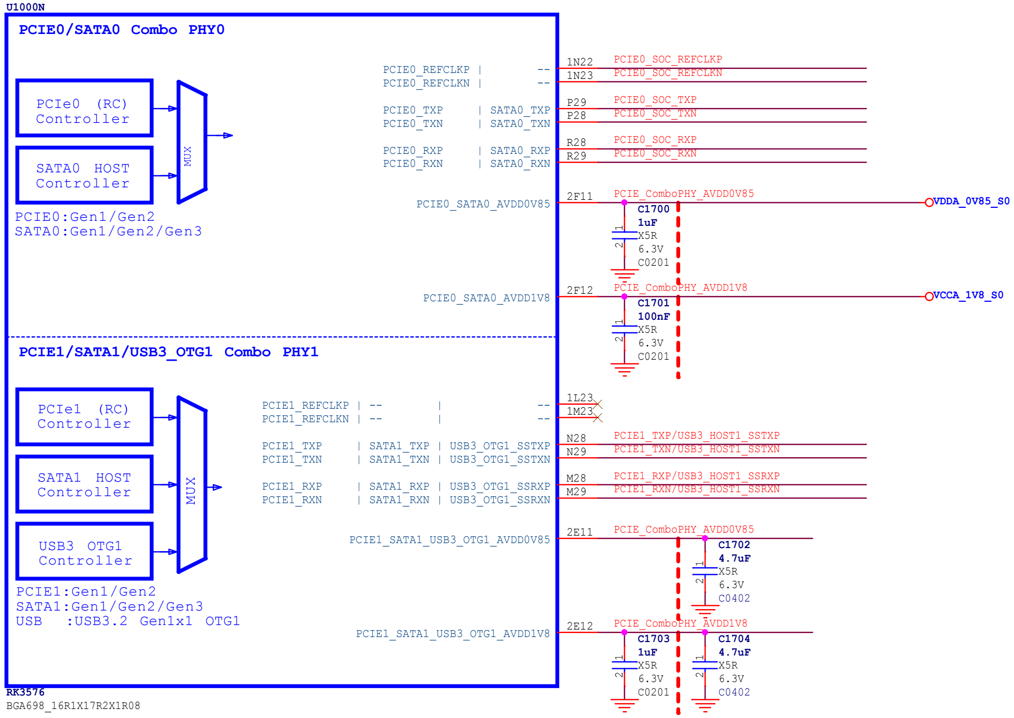
RK3576_N(PCIe/SATA/USB3)



Note!!!
If the PCIe1 or SATA1 function of Combo PHY1 is selected, the USB3 OTG1 function cannot be used, and even the USB2 PHY1 function cannot be used

Note:
Caps of between dashed red lines and U1000 should be placed under the U1000 package. Other caps should be placed close to the U1000 package

深圳市橙视科技发展有限公司		Document Title: K7(RK3576+LPDDR4)	
Page Title: RK3576-PCIe/SATA		PCB NO.: K7_V1.1	
Page Size: A3	Sch Designer: TIM	Pcb Designer: TIM	Project Stage: DR4
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RK3576_F(VCCIO2)

U1000F

VCCIO2 Domain
Operating Voltage=1.8V/3.3V

PWM2_CH5_M0	--	--	--	--	--	AUPLL_CLK_IN_M2	SAI4_MCLK_M0	SAI1_MCLK_M0	GPIO4_A2_d	1D6	GPIO4_A2_d → SAI1_MCLK_M0 [37]
PWM2_CH4_M1	--	I2C2_SCL_M2	UART5_RTSN_M1	SPI3_CSN0_M2	FLEXBUS1_CSN_M4	--	--	SAI1_SCLK_M0	GPIO4_A3_d	1C6	GPIO4_A3_d → SAI1_SCLK_M0 [37]
CAN0_TX_M2	--	I2C4_SCL_M1	UART6_TX_M0	SPI3_MOSI_M2	FLEXBUS0_D13_M1	PDM1_SDI3_M1	SAI4_SCLK_M0	--	GPIO4_A4_d	1C5	GPIO4_A4_d → GPIO4_A4_d [42]
--	PCIE1_CLKREQN_M2	I2C2_SDA_M2	UART5_CTSN_M1	SPI4_CSN1_M2	FLEXBUS1_D12_M1	--	--	SAI1_LRCK_M0	GPIO4_A5_d	1B6	GPIO4_A5_d → SAI1_LRCK_M0 [37]
CAN0_RX_M2	--	I2C4_SDA_M1	UART6_RX_M0	SPI3_MISO_M2	FLEXBUS0_D14_M1	PDM1_CLK0_M1	SAI4_LRCK_M0	--	GPIO4_A6_d	1B5	GPIO4_A6_d → GPIO4_A6_d [42]
PWM2_CH6_M0	--	--	--	SPI3_CLK_M2	--	--	SAI4_SDI_M0	SAI1_SDO0_M0	GPIO4_A7_d	A7	GPIO4_A7_d → SAI1_SDO0_M0 [37]
--	UART2_RTSN_M1	UART6_RTSN_M0	UART5_TX_M1	SPI4_CLK_M2	FLEXBUS1_D13_M1	PDM1_CLK1_M1	SAI1_SDI3_M0	SAI1_SDO1_M0	GPIO4_B0_d	1A5	GPIO4_B0_d → PDM1_CLK1_M1 [42]
--	UART2_CTSN_M1	UART6_CTSN_M0	UART5_RX_M1	SPI4_MOSI_M2	FLEXBUS1_D14_M1	PDM1_SDI2_M1	SAI1_SDI2_M0	SAI1_SDO2_M0	GPIO4_B1_d	B7	GPIO4_B1_d → PDM1_SDI2_M1 [42]
--	--	MIPI_TE_M0	--	SPI4_MISO_M2	FLEXBUS1_D15_M1	PDM1_SDI1_M1	SAI1_SDI1_M0	SAI1_SDO3_M0	GPIO4_B2_d	B6	GPIO4_B2_d → PDM1_SDI1_M1 [42]
PWM2_CH7_M0	--	--	SPI3_CSN1_M2	SPI4_CSN0_M2	--	PDM1_SDI0_M1	SAI4_SDO_M0	SAI1_SDI0_M0	GPIO4_B3_d	1A6	GPIO4_B3_d → R1802 0R R0402 → SAI1_SDI0_M0 [37]
CAN1_RX_M2	--	I2C3_SDA_M0	UART2_RX_M1	--	FLEXBUS0_CSN_M4	--	--	SPDIF_RX0_M0	GPIO4_B4_d	B8	I2C3_SDA_M0_Audio [37]
CAN1_TX_M2	PCIE0_CLKREQN_M2	I2C3_SCL_M0	UART2_TX_M1	--	FLEXBUS0_D15_M1	--	--	SPDIF_TX0_M0	GPIO4_B5_d	1A4	I2C3_SCL_M0_Audio [37]

VCCIO2_VCC 2A2 → VCCIO_AUDIO

C1800 100nF X5R 6.3V C0201

RK3576
BGA698_16R1X17R2X1R08

VCCIO_AUDIO
R1800 2.2K R0201
R1801 2.2K R0201
I2C3_SDA_M0_Audio
I2C3_SCL_M0_Audio

RK3576_G(VCCIO3)

Note:
Caps of between dashed red lines and U1000 should be placed under the U1000 package. Other caps should be placed close to the U1000 package

U1000G

VCCIO3 Domain
Operating Voltage=1.8V/3.3V

PWM1_CH0_M1	PCIE1_CLKREQN_M1	SPI1_CLK_M0	I2C9_SDA_M1	--	--	--	SAI3_SCLK_M1	--	SDMMC1_D0_M0	ETH1_RXD2_M1	GPIO1_B4_d	A28		SDMMC1_D0_M0 [34]
PWM1_CH1_M1	--	SPI1_MOSI_M0	I2C9_SCL_M1	--	--	--	SAI3_LRCK_M1	--	SDMMC1_D1_M0	ETH1_RXD3_M1	GPIO1_B5_d	B27		SDMMC1_D1_M0 [34]
--	PCIE0_CLKREQN_M1	SPI1_MISO_M0	--	UART3_CTSN_M2	--	--	SAI3_SDO_M1	--	SDMMC1_D2_M0	ETH1_RXCLK_M1	GPIO1_B6_d	1A23		SDMMC1_D2_M0 [34]
--	--	SPI1_CSN0_M0	--	UART3_RTSN_M2	--	--	SAI3_SDI_M1	--	SDMMC1_D3_M0	ETH1_TXD2_M1	GPIO1_B7_d	A27		SDMMC1_D3_M0 [34]
PWM0_CH0_M1	--	SPI1_CSN1_M0	--	UART3_TX_M2	--	PDM0_SDI2_M2	--	--	SDMMC1_CMD_M0	ETH1_TXD3_M1	GPIO1_C0_d	B26		SDMMC1_CMD_M0 [34]
--	--	--	--	UART3_RX_M2	--	PDM0_CLK0_M2	SAI3_MCLK_M1	--	SDMMC1_CLK_M0	ETH1_TXCLK_M1	GPIO1_C1_d	1B22	GPIO1_C1_d	SDMMC1_CLK_M0 [34]
PWM1_CH2_M1	--	SPI2_CSN1_M1	I2C6_SCL_M1	UART4_RTSN_M1	--	--	FSPI1_CSN1_M1	FSPI1_RSTN_M1	SDMMC1_PWREN_M0	ETH1_PPSCLK_M1	GPIO1_C2_u	B29	GPIO1_C2_u	UART4_RTSN_M1 [34]
--	--	SPI2_CSN0_M1	I2C6_SDA_M1	UART4_CTSN_M1	--	--	--	FSPI1_CSN0_M1	SDMMC1_DETN_M0	ETH1_PPSTRIG_M1	GPIO1_C3_u	1C23	GPIO1_C3_u	UART4_CTSN_M1 [34]
--	PCIE0_BUTTONRSTN	SPI2_MOSI_M1	UART2_RTSN_M0	UART4_TX_M1	--	--	--	FSPI1_D0_M1	--	ETH1_TXD0_M1	GPIO1_C4_d	1B23	GPIO1_C4_d	UART4_TX_M1 [34]
--	PCIE1_BUTTONRSTN	SPI2_MISO_M1	UART2_CTSN_M0	UART4_RX_M1	--	--	--	FSPI1_D1_M1	--	ETH1_TXD1_M1	GPIO1_C5_d	B28	GPIO1_C5_d	UART4_RX_M1 [34]
--	SATA_CPPOD	--	I2C8_SCL_M1	UART2_TX_M0	--	PDM0_SDI0_M2	--	FSPI1_D2_M1	--	ETH1_TXCTL_M1	GPIO1_C6_d	A26	GPIO1_C6_d	WIFI_REG_ON_H [34]
--	SATA_CPDET	--	I2C8_SDA_M1	UART2_RX_M0	--	PDM0_SDI1_M2	--	FSPI1_D3_M1	--	ETH1_RXD0_M1	GPIO1_C7_d	1C22	GPIO1_C7_d	BT_REG_ON_H [34]
--	--	--	--	UART10_TX_M1	--	--	SAI2_SDO_M0	--	--	ETH1_RXD1_M1	GPIO1_D0_d	C29	GPIO1_D0_d	UART10_TX_M1 [42]
--	--	--	I3C0_SDA_PU_M1	UART10_RX_M1	--	--	SAI2_SCLK_M0	--	--	ETH1_RXCTL_M1	GPIO1_D1_d	1D22	GPIO1_D1_d	UART10_RX_M1 [42]
PWM1_CH3_M1	--	--	I3C0_SCL_M1	--	--	--	SAI2_LRCK_M0	--	--	ETH1_MDC_M1	GPIO1_D2_d	1A24	GPIO1_D2_d	I3C0_SCL_M1 [42]
PWM1_CH4_M1	--	--	I3C0_SDA_M1	--	--	--	SAI2_SDI_M0	--	--	ETH1_MDIO_M1	GPIO1_D3_d	C28	GPIO1_D3_d	I3C0_SDA_M1 [42]
--	--	--	I2C5_SCL_M1	UART10_RTSN_M1	SPDIF_RX1_M2	PDM0_SDI3_M2	SAI2_MCLK_M0	--	--	ETH1_MCLK_M1	GPIO1_D4_d	1E21	GPIO1_D4_d	HOST_WAKE_BT_H [34]
CLK1_32K_OUT	SATA_MPSWIT	SPI2_CLK_M1	I2C5_SDA_M1	UART10_CTSN_M1	SPDIF_TX1_M2	PDM0_CLK1_M2	--	FSPI1_CLK_M1	--	ETH_CLK1_25M_OUT_M1	GPIO1_D5_d	1E22	GPIO1_D5_d	WIFI_WAKE_HOST_H [34]

For BT

For BT

VCCIO3_VCC 2B10 → VCCIO_WL

1.8V_S3

C1801 100nF X5R 6.3V C0201

RK3576
BGA698_16R1X17R2X1R08

RK3576_J(VCCIO6)

U1000J

VCCIO6 Domain
Operating Voltage=1.8V/3.3V

PWM1_CH5_M1	UART11_TX_M2	SPI4_CSN1_M0	I2C7_SCL_M3	--	HDMI_TX_CEC_M0	--	SAI4_MCLK_M2	DSM_AUD_LP_M1	GPIO4_C0_d	AK3	GPIO4_C0_d
PWM0_CH1_M1	UART11_RX_M2	EDP_TX_HPDIN_M0	I2C7_SDA_M3	PCIE1_CLKREQN_M3	HDMI_TX_HPDIN_M0	--	--	DSM_AUD_LN_M1	GPIO4_C1_d	AK2	HDMI_TX_HPDIN_M0 [32]
PWM2_CH0_M1	UART9_TX_M2	CAN0_TX_M1	I2C2_SCL_M3	--	HDMI_TX_SCL	--	--	DSM_AUD_RP_M1	GPIO4_C2_d	AL2	GPIO4_C2_d
PWM2_CH1_M1	UART9_RX_M2	CAN0_RX_M1	I2C2_SDA_M3	--	HDMI_TX_SDA	--	--	DSM_AUD_RN_M1	GPIO4_C3_d	1AE2	GPIO4_C3_d
PWM2_CH6_M1	UART6_TX_M3	SPI4_CSN0_M0	I2C3_SCL_M3	DP_HPDIN_M0	--	--	SAI4_LRCK_M2	ISP_PRELIGHT_TRIG_M1	GPIO4_C4_d	AL3	GPIO4_C4_d
PWM2_CH5_M1	UART6_RX_M3	SPI4_MOSI_M0	I2C3_SDA_M3	SATA1_ACTLED_M1	--	VP0_SYNC_OUT	SAI4_SDO_M2	ISP_FLASH_TRIGOUT_M1	GPIO4_C5_d	AK1	GPIO4_C5_d
PWM2_CH2_M1	CAN1_TX_M1	SPI4_MISO_M0	I2C6_SCL_M3	SATA0_ACTLED_M1	PCIE0_CLKREQN_M3	VP1_SYNC_OUT	SAI4_SDI_M2	--	GPIO4_C6_d	1AE1	GPIO4_C6_d
PWM2_CH3_M1	CAN1_RX_M1	SPI4_CLK_M0	I2C6_SDA_M3	--	--	VP2_SYNC_OUT	SAI4_SCLK_M2	--	GPIO4_C7_d	AJ1	GPIO4_C7_d

GPIO4_C0_d → HDMI_TX_CEC_M0 [32]
GPIO4_C2_d → HDMI_TX_SCL [32]
GPIO4_C3_d → HDMI_TX_SDA [32]
GPIO4_C5_d → GPIO4_C5_d [29]
GPIO4_C4_d → USB_OTG0_PWREN_H [23]
GPIO4_C6_d → MIPI_DPHY_CSI1_PDN_H [29]
GPIO4_C7_d → MIPI_DPHY_CSI0_PWM [29]

VCCIO6_VCC 2N3 → VCC_3V3_S0

C1802 100nF X5R 6.3V C0201

RK3576
BGA698_16R1X17R2X1R08

Default turn off
Option: For test

Oranth 深圳市橙视科技发展有限公司	Document Title: K7(RK3576+LPDDR4)		
Page Title: RK3576-GPIO VCCIO2	PCB NO.: K7_V1.1		
Page Size: A3	Sch Designer: TIM	Pcb Designer: TIM	Project Stage: DR4
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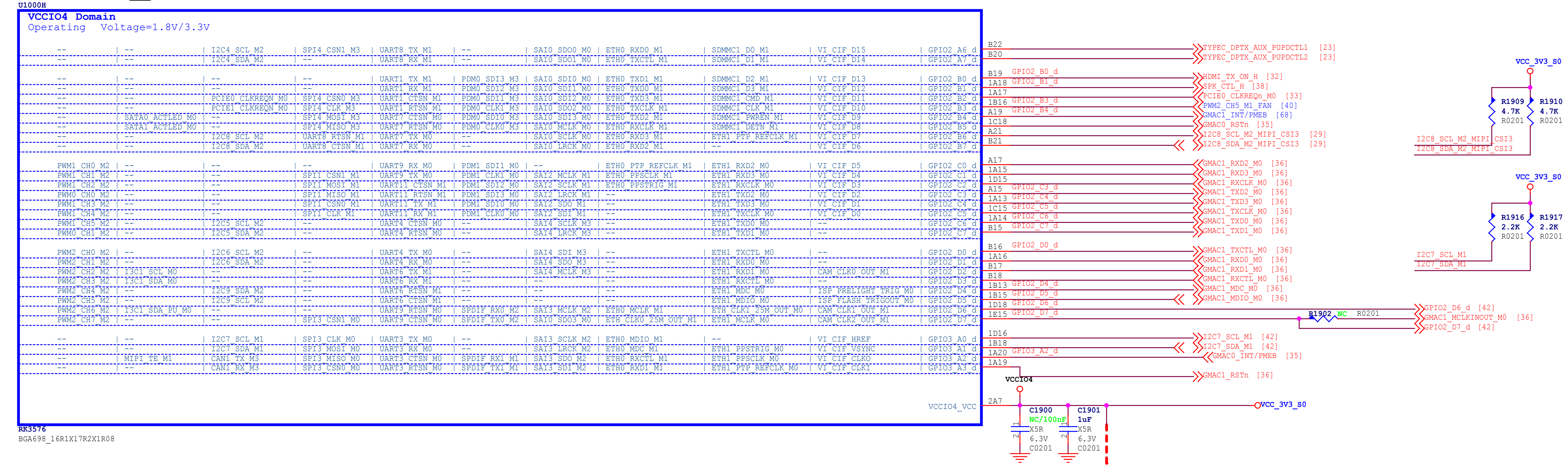
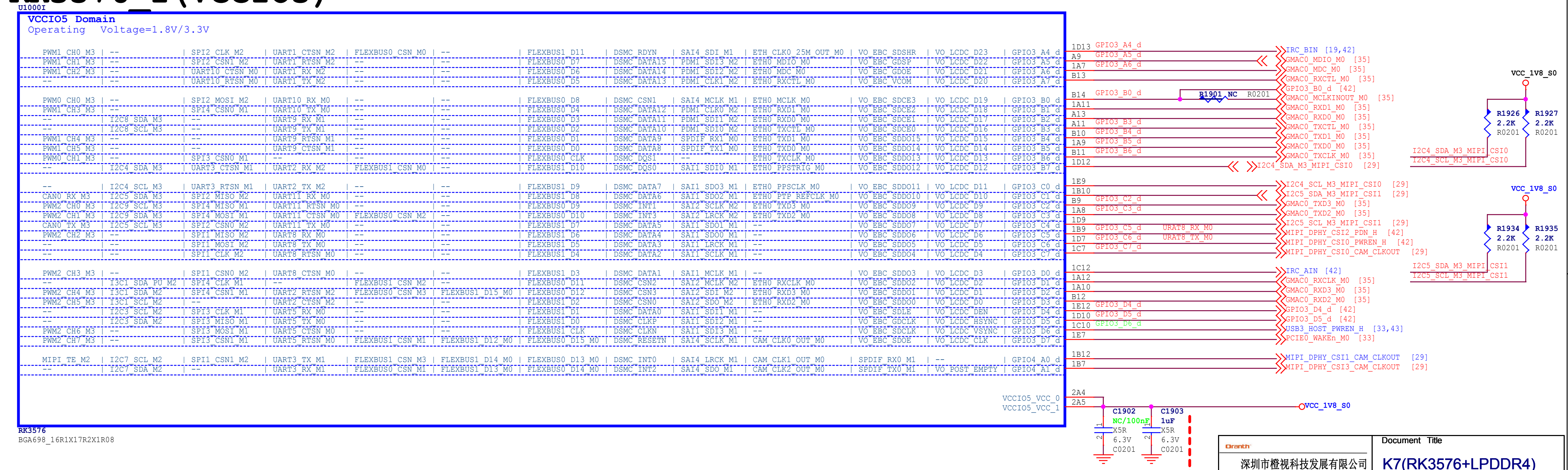
RK3576_H(VCCIO4)



RK3576_I(VCCIO5)



Orenth 深圳市橙视科技发展有限公司	Document Title: K7(RK3576+LPDDR4)	
Page Title: RK3576-GPIO VCCIO4/5	PCB NO.: K7_V1.1	
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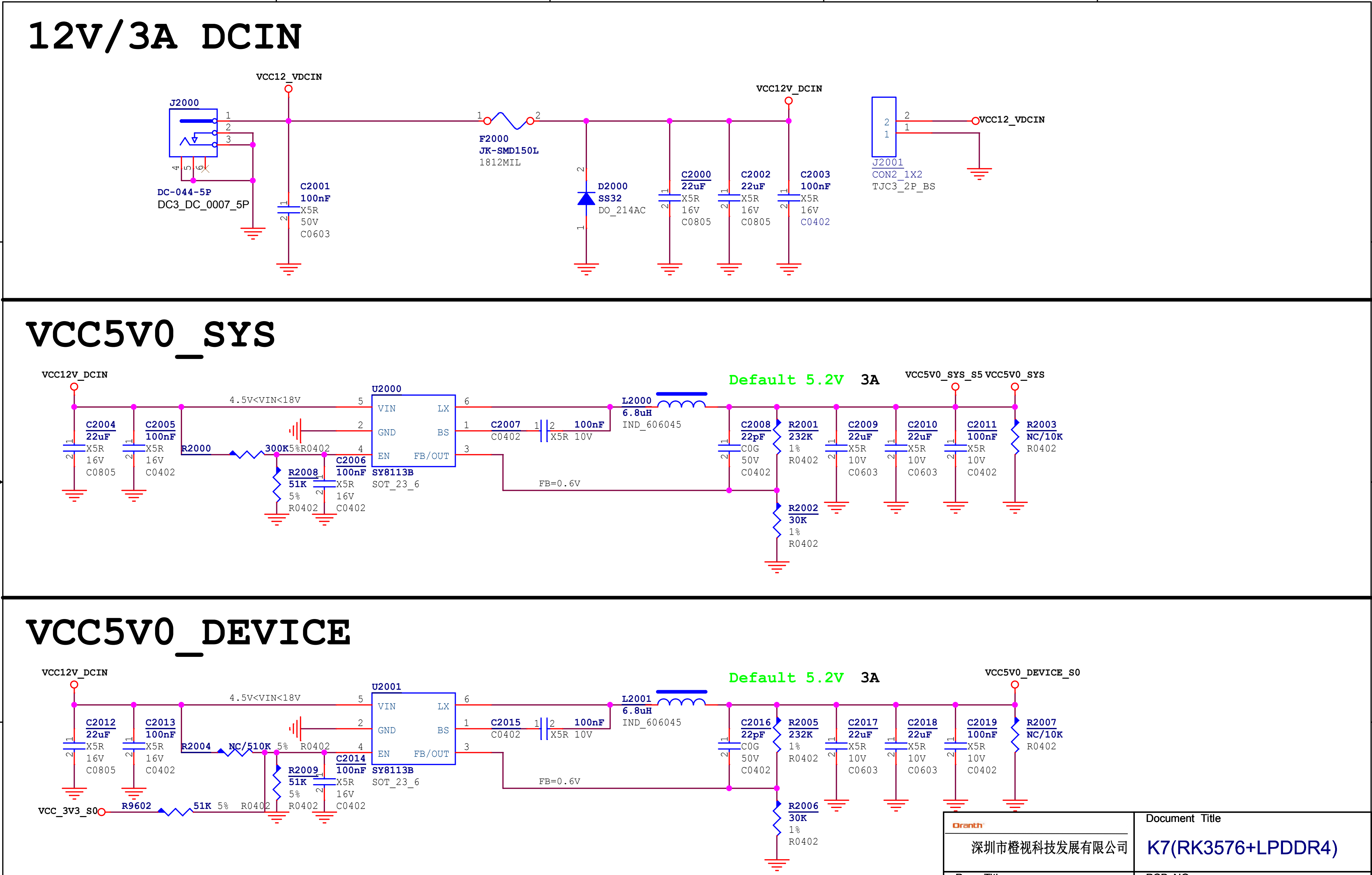

深圳市橙视科技发展有限公司	Document Title: K7(RK3576+LPDDR4)
Page Title: Power-DC IN	PCB NO.: K7_V1.1

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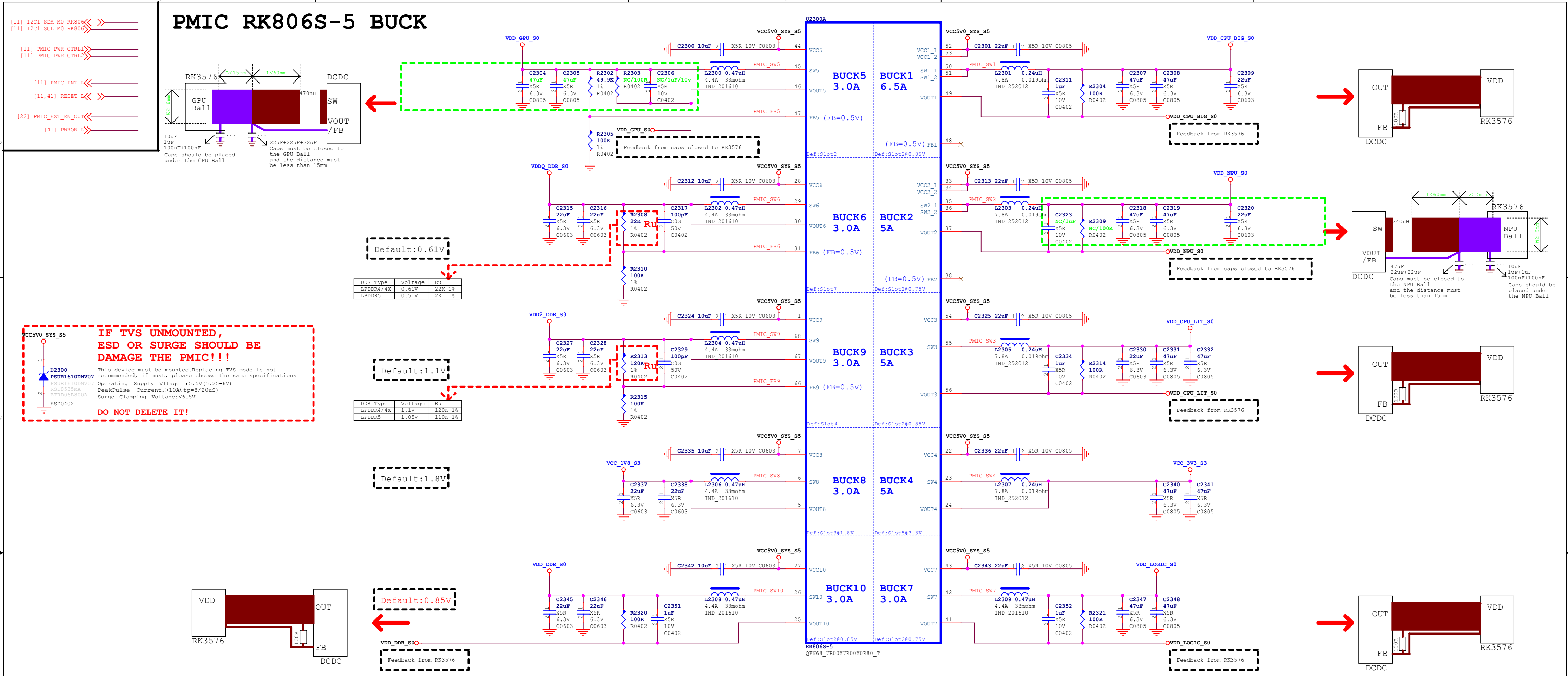
PMIC RK806S-5 BUCK



DDR Type	Voltage	Ru
LPDDR4/4X	0.61V	22K 1%
LPDDR5	0.51V	2K 1%

DDR Type	Voltage	Ru
LPDDR4/4X	1.1V	120K 1%
LPDDR5	1.05V	110K 1%

VCC5V0_SYS_S5

D2300
PSUR1610DNV07
PSUR1610DNV07
RSD8535MA
BTRD06B800A
ESD0402

IF TVS UNMOUNTED, ESD OR SURGE SHOULD BE DAMAGE THE PMIC!!!

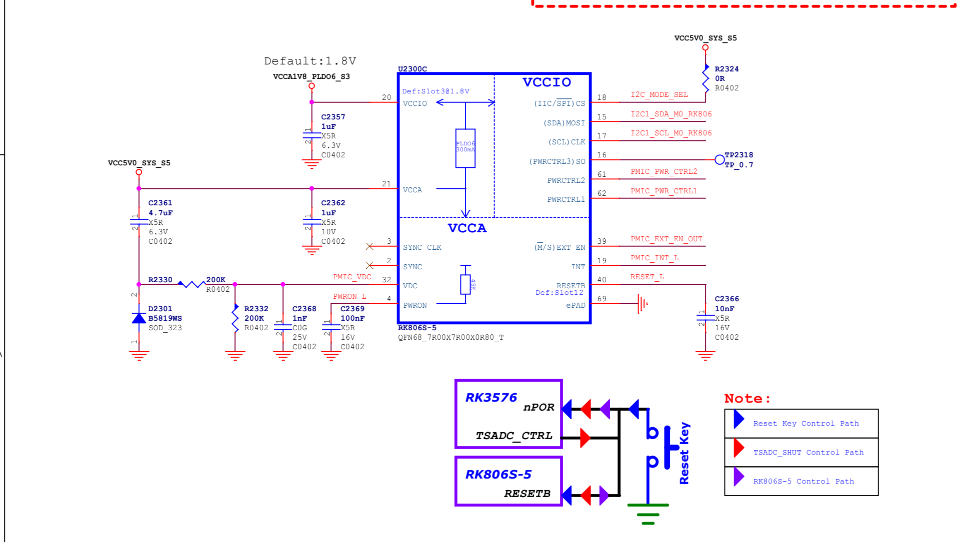
This device must be mounted.Replacing TVS mode is not recommended, if must, please choose the same specifications

Operating Supply Vltage :5.5V(5.25-6V)
PeakPulse Current:>10A(tp=8/20uS)
Surge Clamping Voltage:<6.5V

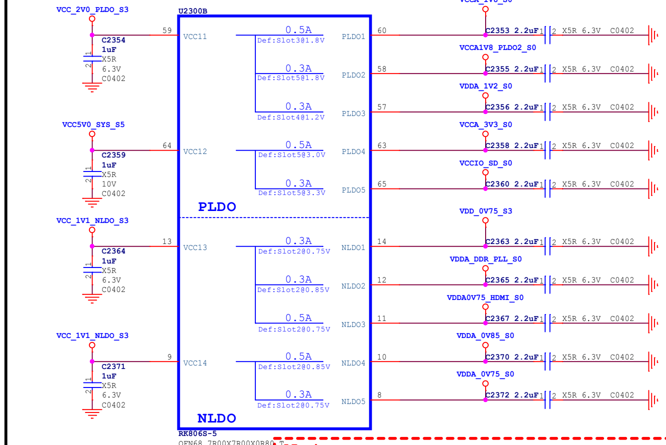
DO NOT DELETE IT!

PMIC RK806S-5 Managerment

Note:
I2C Mode:CS(pin18) connected to VCCA(pin21);
SPI Mode(Def):CS(pin18) floating or connected to GND



PMIC RK806S-5 LDO



Default:1.8V

Default:1.8V

Default:1.2V

Default:3.0V

Default:3.3V

Default:0.75V

Default:0.85V

Default:0.75V

Default:0.85V

Default:0.75V

Note:

The RK806 LDO power distribution of the reference chematics is only suitable for the interface used in the reference schematics.
If other interface functions are to be added to the reference schematics, the RK806 LDO distribution needs to be re evaluated, otherwise the added functions may exceed the maximum current provided by the LDO

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Page Title: Power-PMIC RK806S-5	PCB NO.: K7_V1.1		
Page Size: A2	Sch Designer: TIM	Pcb Designer: TIM	Project Stage: DR4
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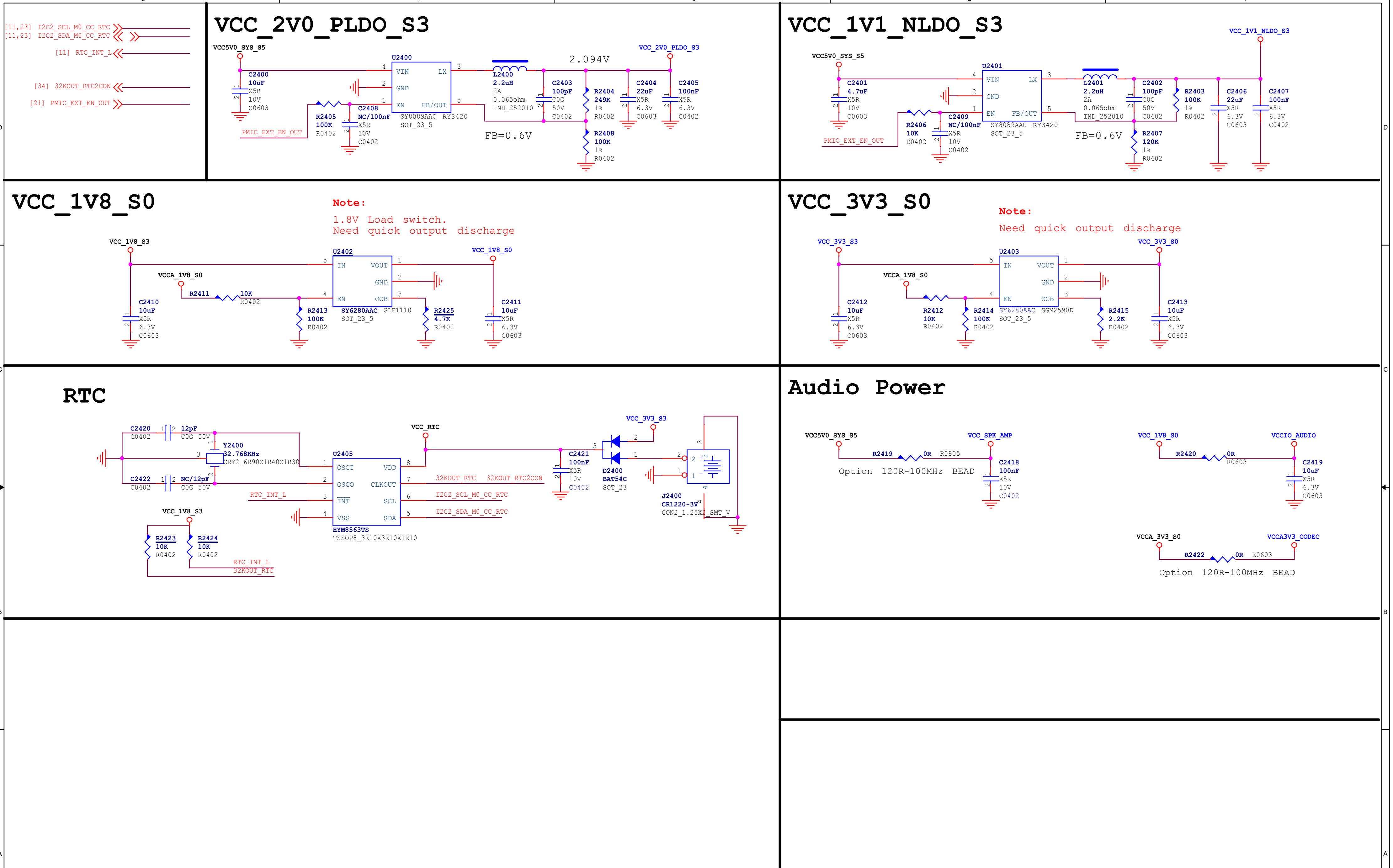

深圳市橙视科技发展有限公司	Document Title: K7(RK3576+LPDDR4)
Page Title: Power-Ext Discrete/RTC	PCB NO.: K7_V1.1
Page Size: A3	Sch Designer: TIM
Pcb Designer: TIM	Project Stage: DR4
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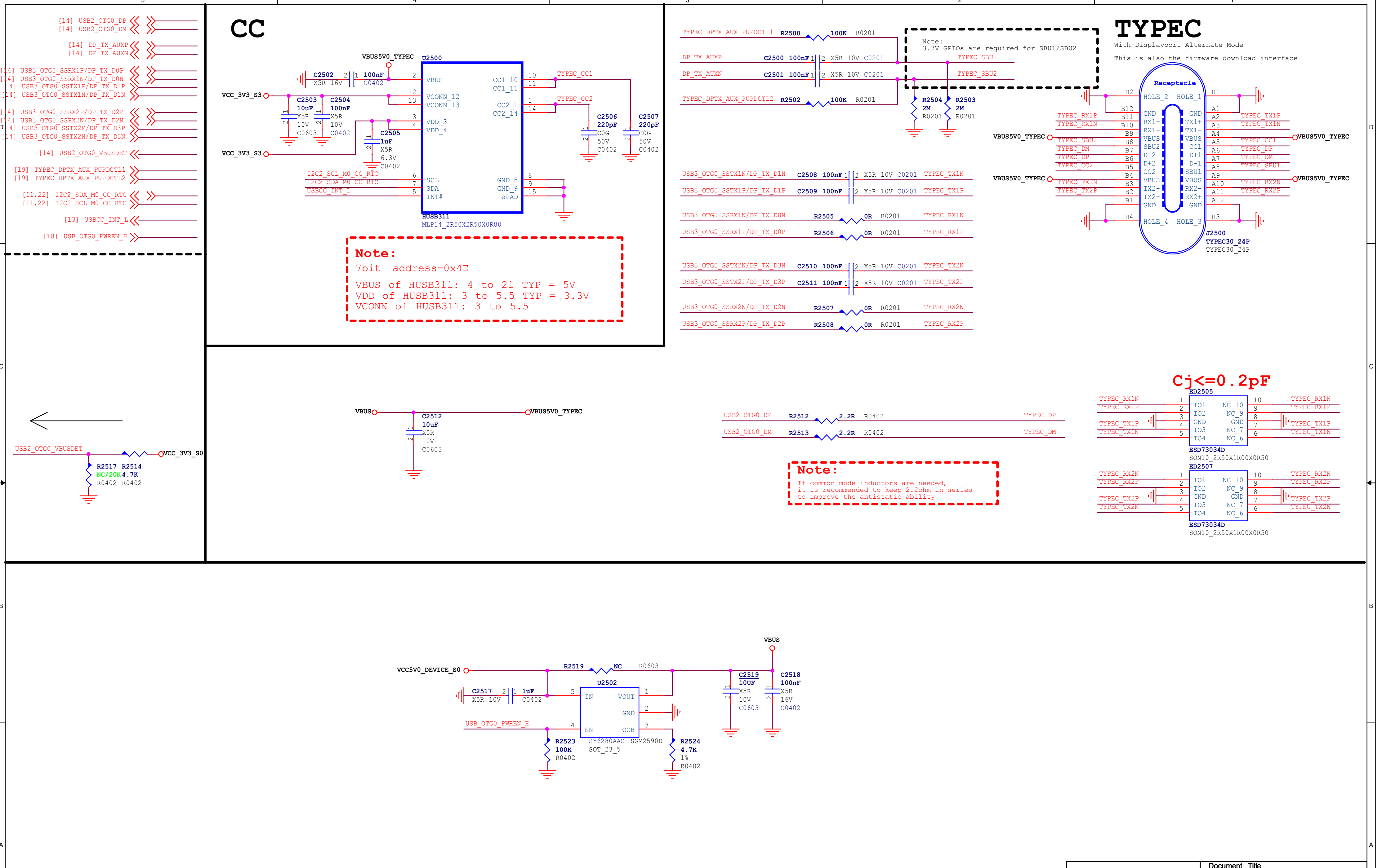

深圳市橙视科技发展有限公司		Document Title	K7(RK3576+LPDDR4)
Page Title	USB/TypeC Port	PCB NO.	K7_V1.1
Page Size: A3	Sch Designer: TIM	Pcb Designer: TIM	Project Stage: DR4
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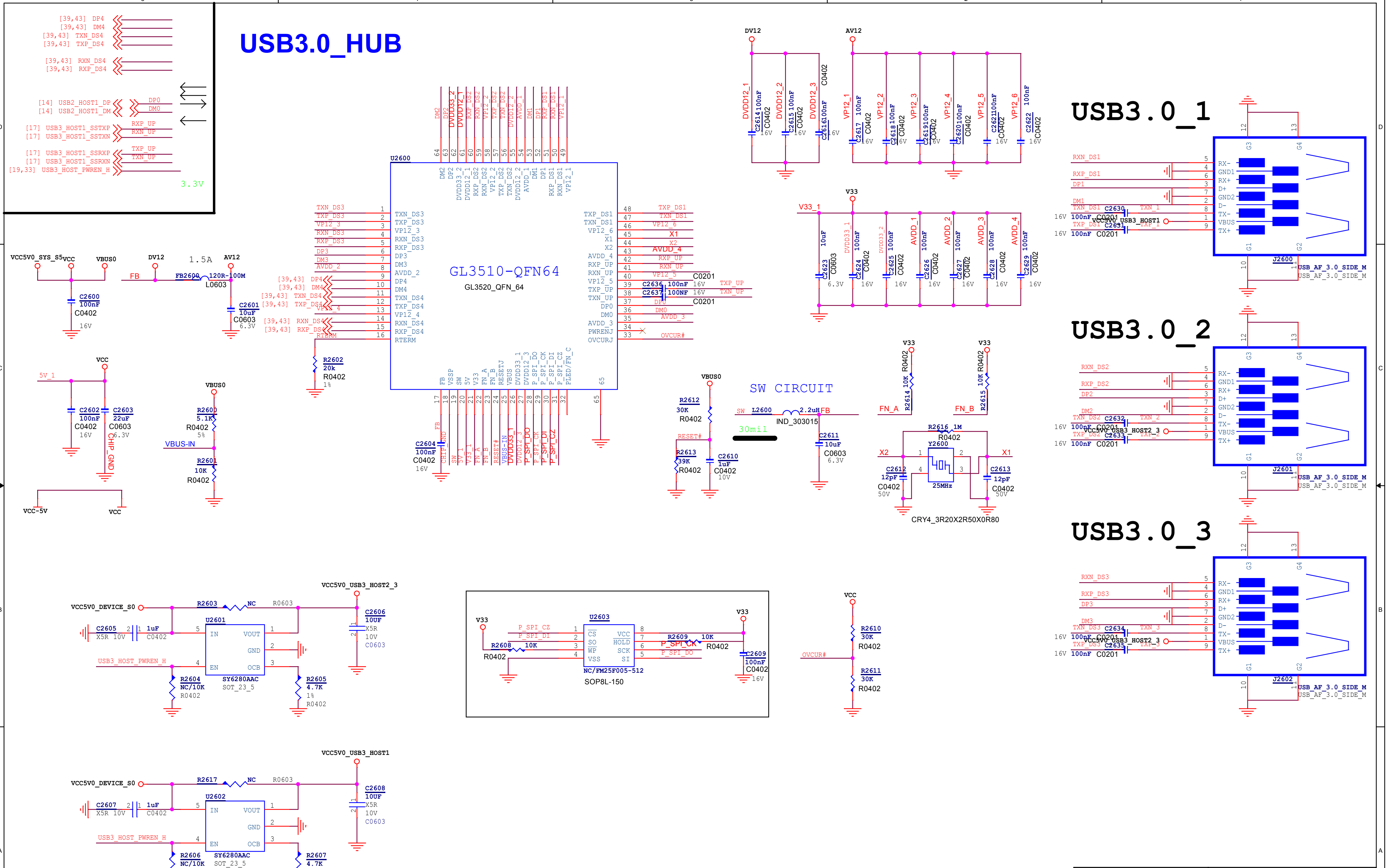

深圳市橙视科技发展有限公司	Document Title	K7(RK3576+LPDDR4)
Page Title: USB30-HUB	PCB NO.	K7_V1.1
Page Size: A3	Sch Designer: TIM	Pcb Designer: TIM
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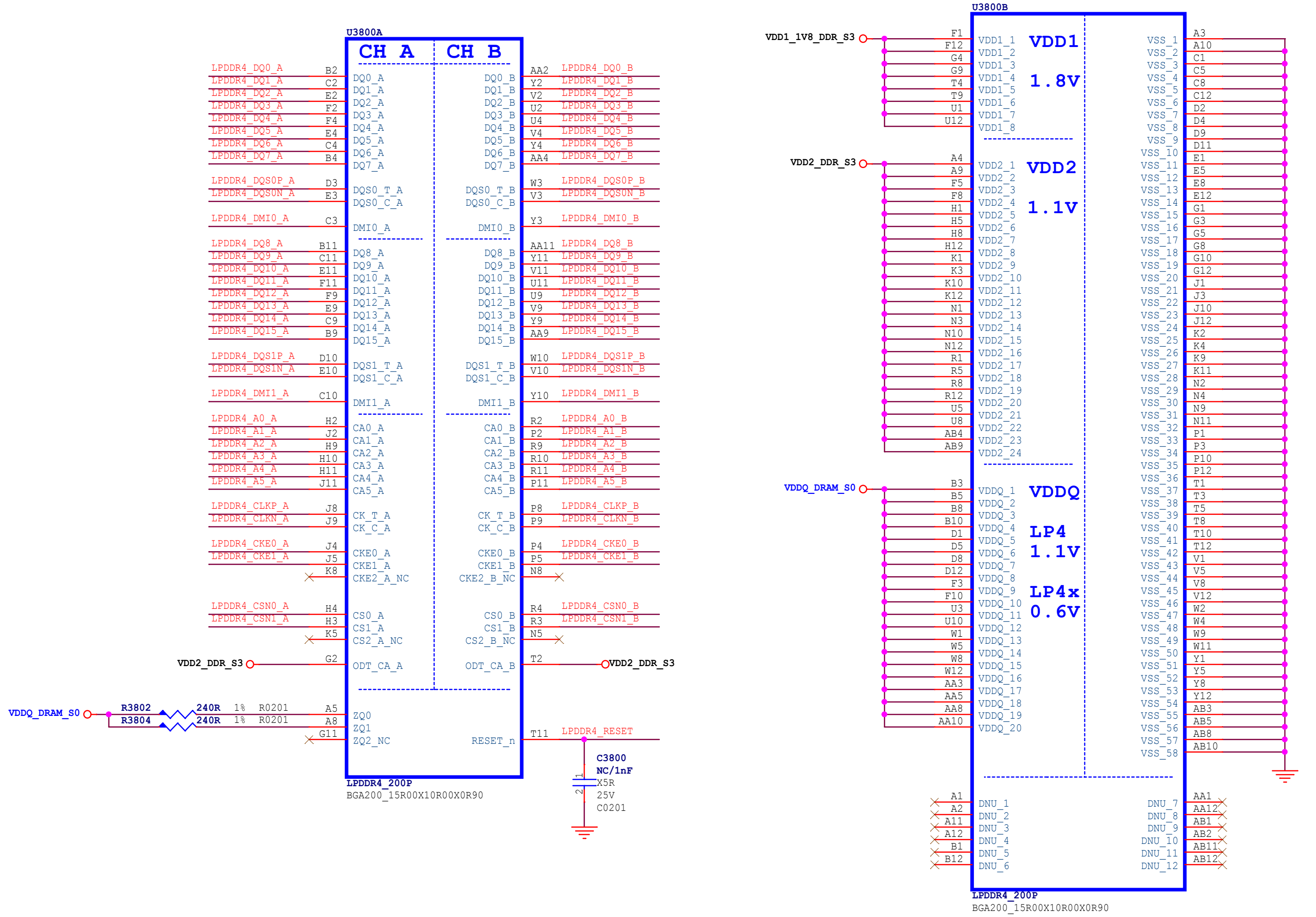
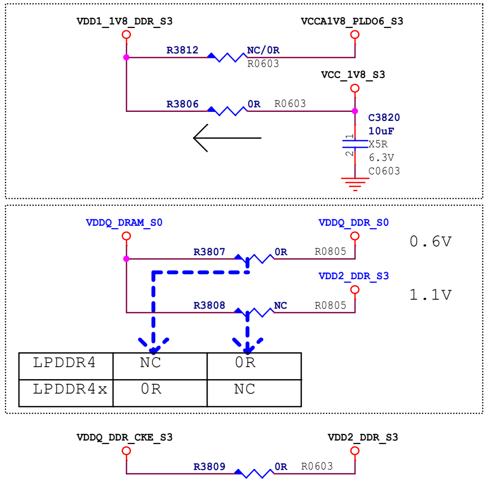
LPDDR4/4X



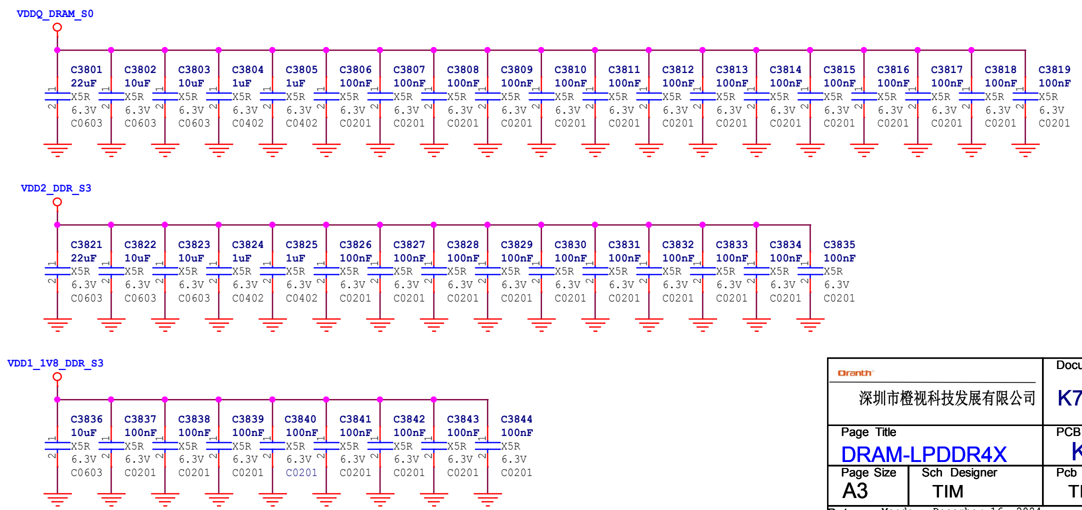
Note:

Sequence:VDD1-VDD2-VDDQ

	LPDDR4	LPDDR4X
VDD1:	1.70-1.95	1.70-1.95
VDD2:	1.06-1.17	1.06-1.17
VDDQ:	1.06-1.17	0.57-0.65



LPDDR4	NC	0R
LPDDR4x	0R	NC



深圳市橙视科技发展有限公司	Document Title: K7(RK3576+LPDDR4)
Page Title: DRAM-LPDDR4X	PCB NO.: K7_V1.1
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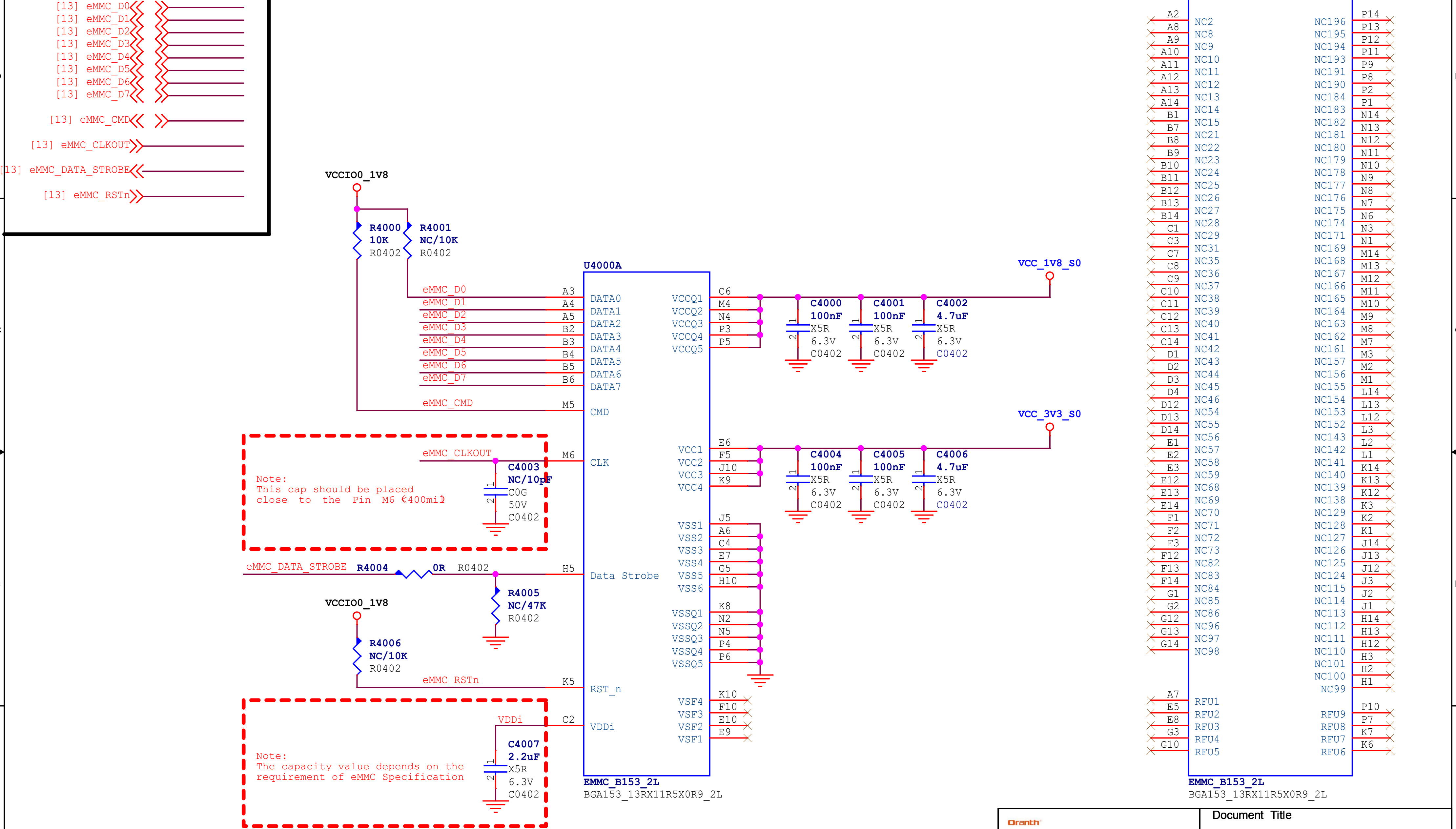
eMMC FLASH



Oranth 深圳市橙视科技发展有限公司	Document Title: K7(RK3576+LPDDR4)
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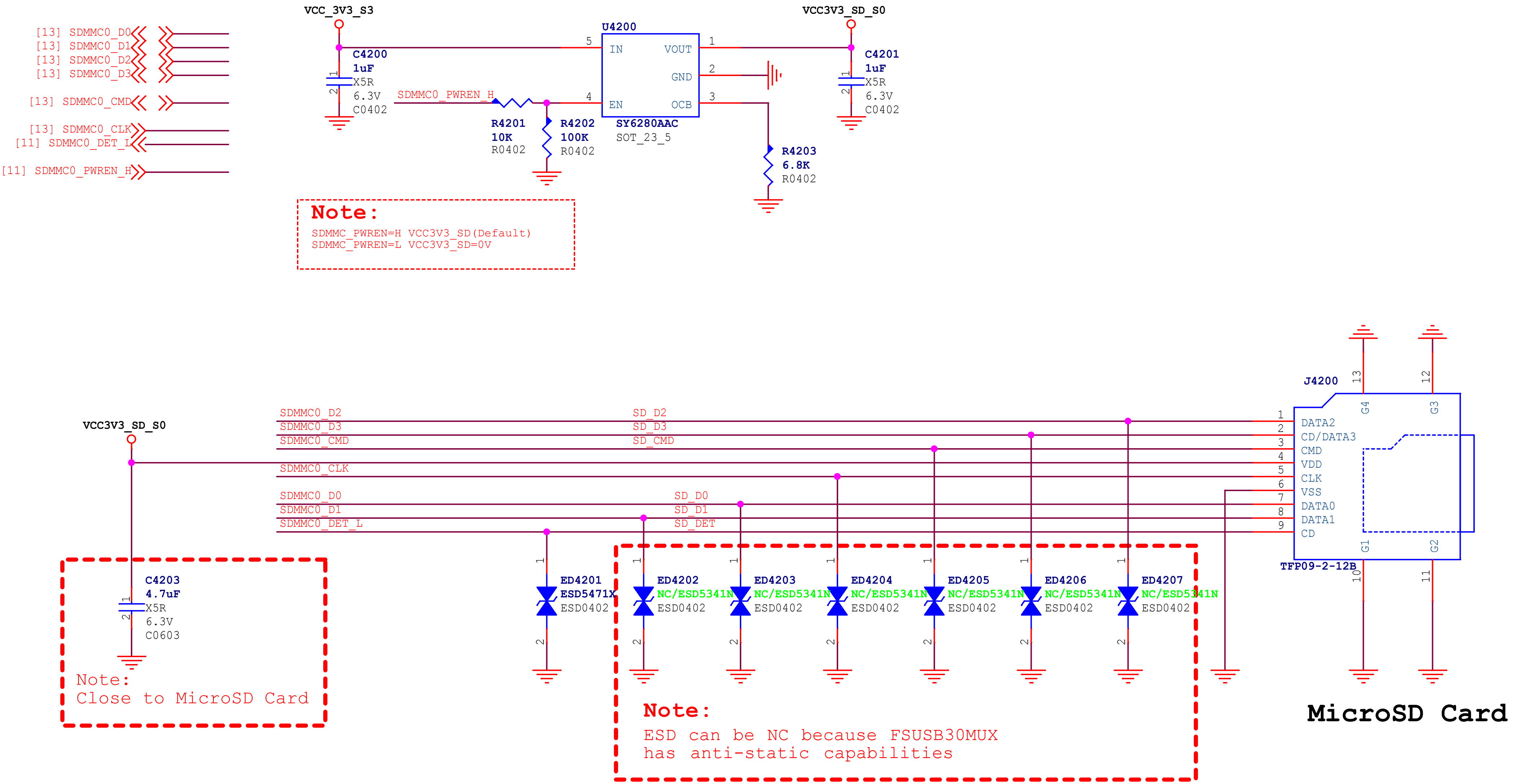
TF CARD



Note:

SDMMC_DET_L:
SDCARD PLUG: Pull-down to GND
SDCARD UNPLUG: Pull-up to PMUIO0_VCC1V8.

Orangepi 深圳市橙视科技发展有限公司		Document Title: K7(RK3576+LPDDR4)	
Page Title: Flash-MicroSD Card		PCB NO.: K7_V1.1	
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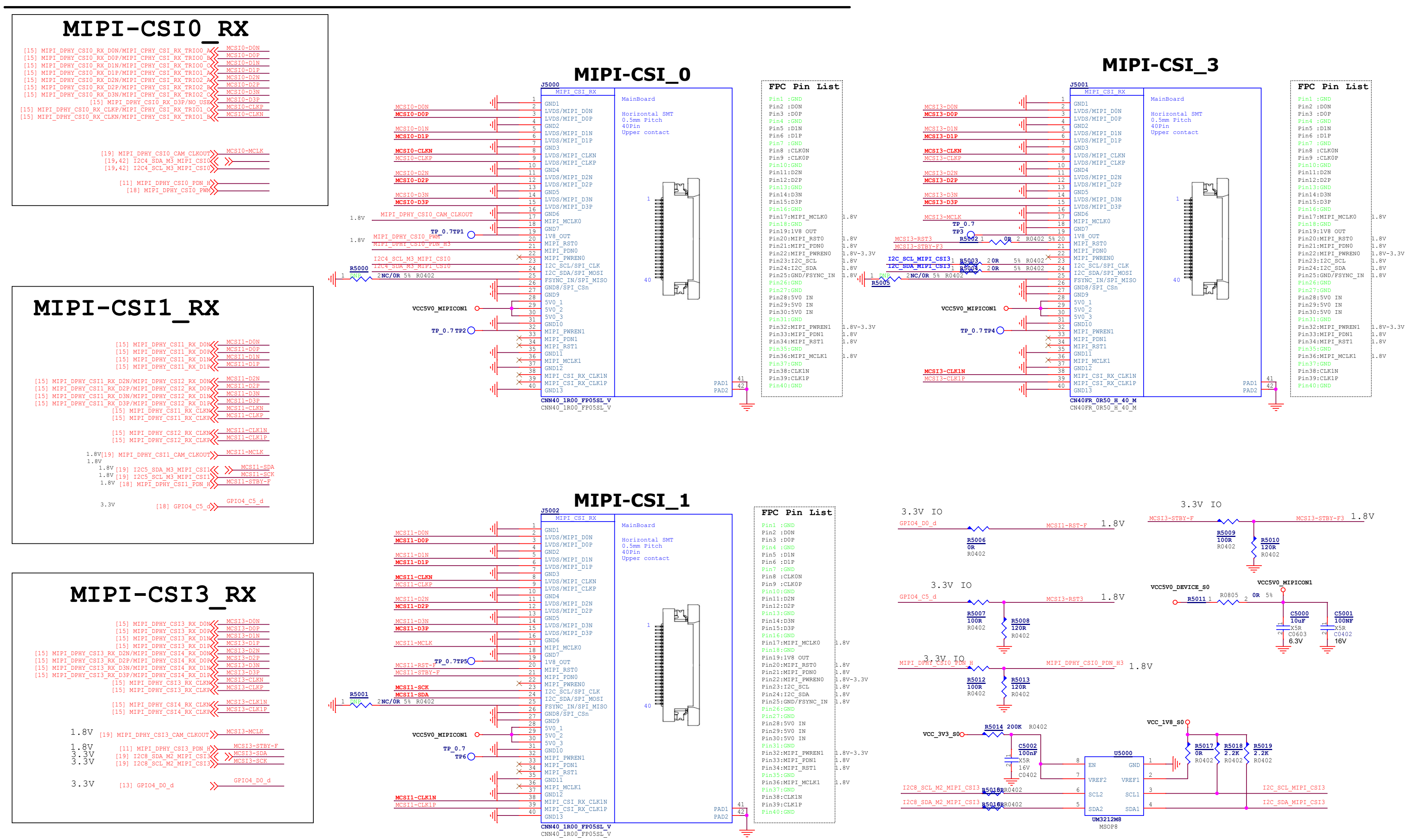

Orenth 深圳市橙视科技发展有限公司	Document Title: K7(RK3576+LPDDR4)
Page Title: VI-Camera_MIPI-DCPHY	PCB NO.: K7_V1.1
Page Size: A2 · Sch Designer: TIM	Pcb Designer: TIM · Project Stage: DR4
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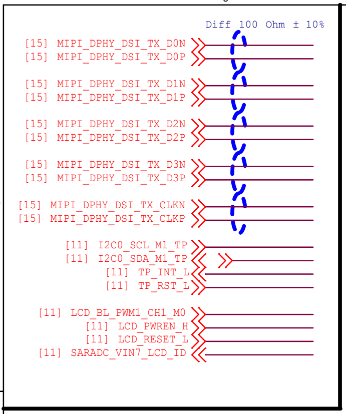
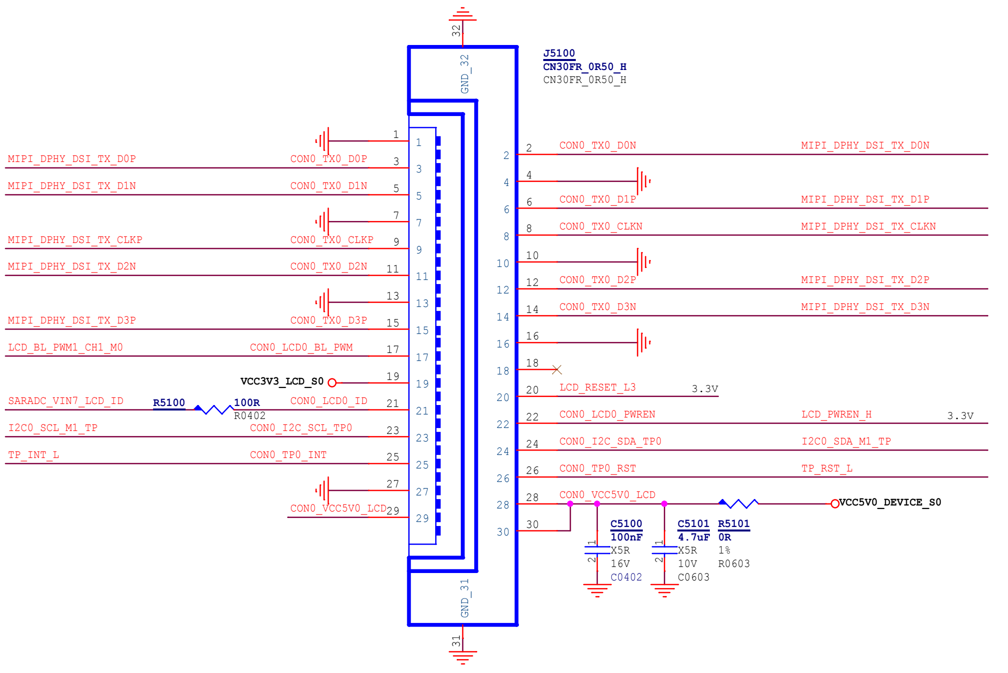
Single-MIPI LCM



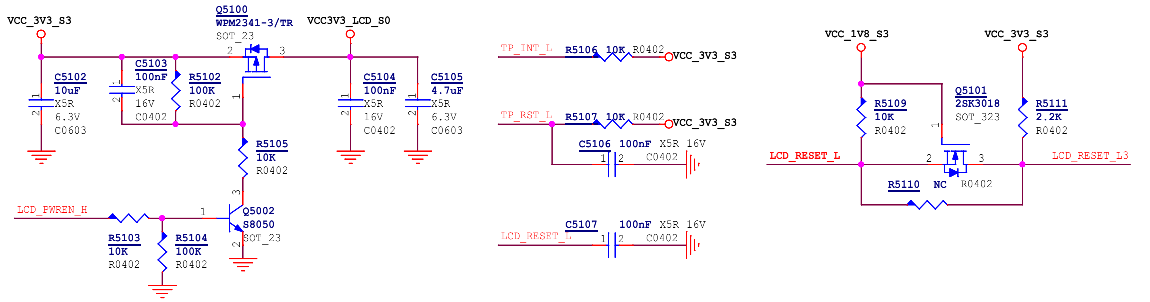


Note:

Signal	IO Level
I2C_SCL_TP	3.3V
I2C_SDA_TP	3.3V
TP_INT_L	3.3V
TP_RST_L	1080P:3.3V 720P:1.8V
LCD_BL_PWM	3.3V
LCD_PWREN_H	1.8V/3.3V
LCD_RST_L	1.8V
MIPI_TE	1.8V
SARADC_VIN7_LCD_ID	1.8V

FPC Pin List

Pin1 :GND
Pin2 :D0N
Pin3 :D0P
Pin4 :GND
Pin5 :D1N
Pin6 :D1P
Pin7 :GND
Pin8 :CLKN/AUXN
Pin9 :CLKP/AUXP
Pin10:GND
Pin11:D2N
Pin12:D2P
Pin13:GND
Pin14:D3N
Pin15:D3P
Pin16:GND
Pin17:LCD_PWM_BL
Pin18:LCD_TE
Pin19:VCC3V3_LCD
Pin20:LCD_RST
Pin21:LCD_ID
Pin22:LCD_PWREN
Pin23:TP_I2C_SCL
Pin24:TP_I2C_SDA
Pin25:TP_INT
Pin26:TP_RST
Pin27:GND
Pin28:5V0
Pin29:5V0
Pin30:5V0



深圳市橙视科技发展有限公司	Document Title: K7(RK3576+LPDDR4)
Page Title: VO-LCM MIPI DPHY TX	PCB NO.: K7_V1.1
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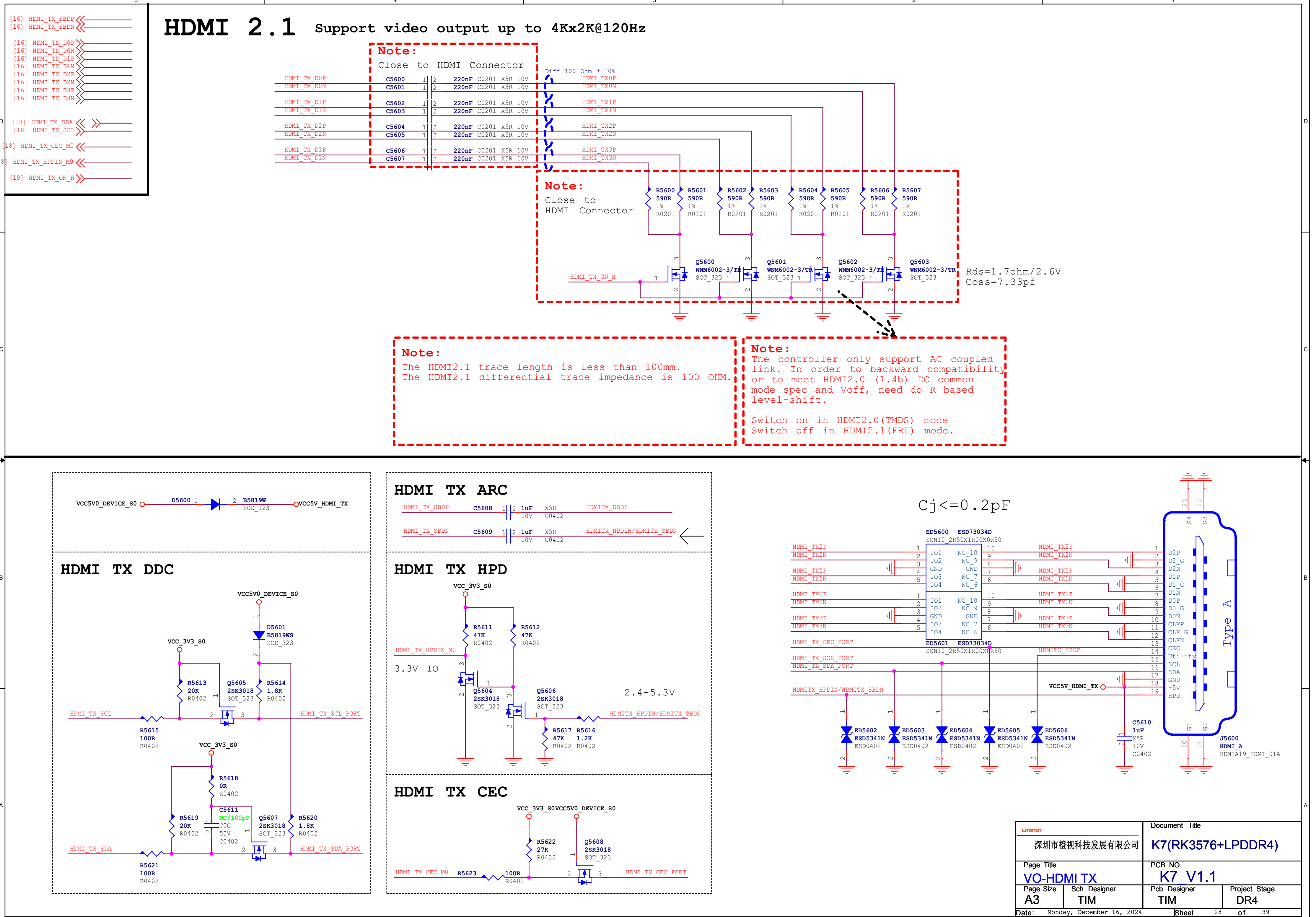
HDMI 2.1 Support video output up to 4Kx2K@120Hz



Note:
The HDMI2.1 trace length is less than 100mm.
The HDMI2.1 differential trace impedance is 100 OHM.

Note:
The controller only support AC coupled link. In order to backward compatibility or to meet HDMI2.0 (1.4b) DC common mode spec and Voff, need do R based level-shift.

Switch on in HDMI2.0(TMDS) mode
Switch off in HDMI2.1(FRL) mode.

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Page Title: VO-HDMI TX	PCB NO.: K7_V1.1
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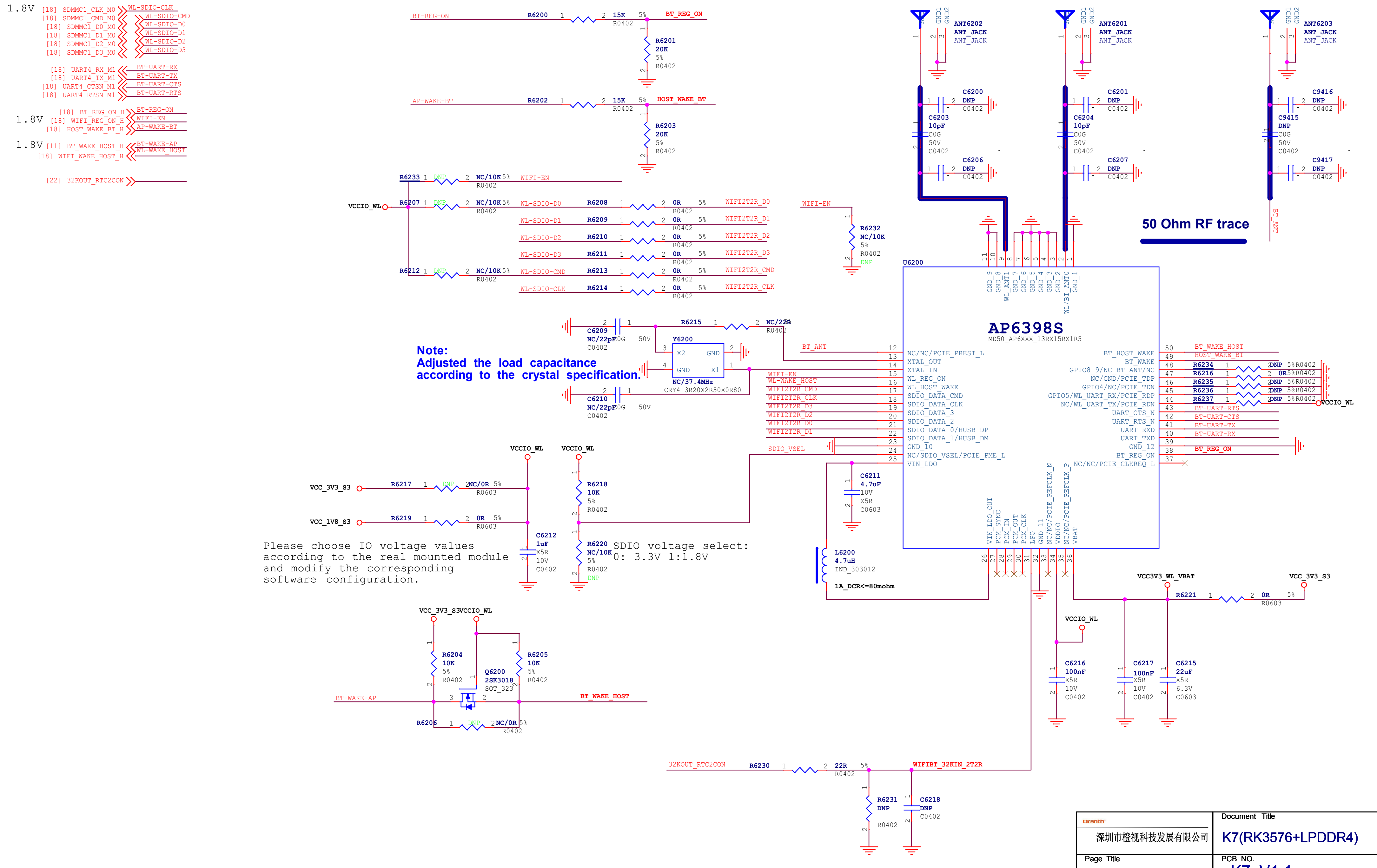

Document Title: K7(RK3576+LPDDR4)

深圳市橙视科技发展有限公司

Page Title	PCB NO.
WIFI/BT-RTL8822	K7_V1.1

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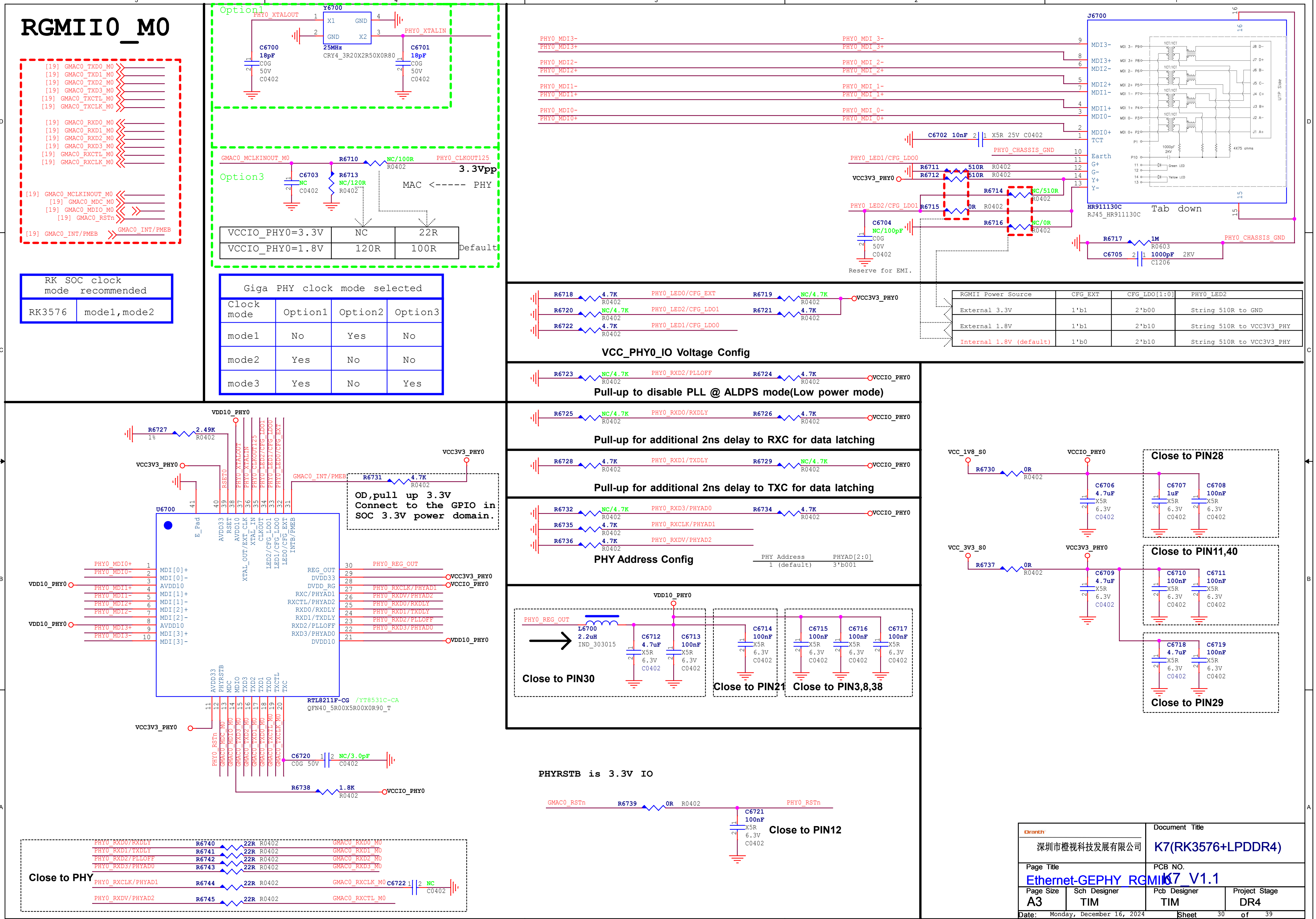
RGMII0_M0



RK SOC clock mode recommended	
RK3576	mode1,mode2

VCCIO_PHY0=3.3V	NC	22R
VCCIO_PHY0=1.8V	120R	100R

Default

Giga PHY clock mode selected

Clock mode	Option1	Option2	Option3
mode1	No	Yes	No
mode2	Yes	No	No
mode3	Yes	No	Yes

RGMII Power Source	CFG_EXT	CFG_LDO[1:0]	PHY0_LED2
External 3.3V	1'b1	2'b00	String 510R to GND
External 1.8V	1'b1	2'b10	String 510R to VCC3V3_PHY
Internal 1.8V (default)	1'b0	2'b10	String 510R to VCC3V3_PHY

PHY Address	PHYAD[2:0]
1 (default)	3'b001

深圳市橙视科技发展有限公司	Document Title: K7(RK3576+LPDDR4)		
Page Title: Ethernet-GEPHY RGMII0	PCB NO.: K7_V1.1		
Page Size: A3	Sch Designer: TIM	Pcb Designer: TIM	Project Stage: DR4
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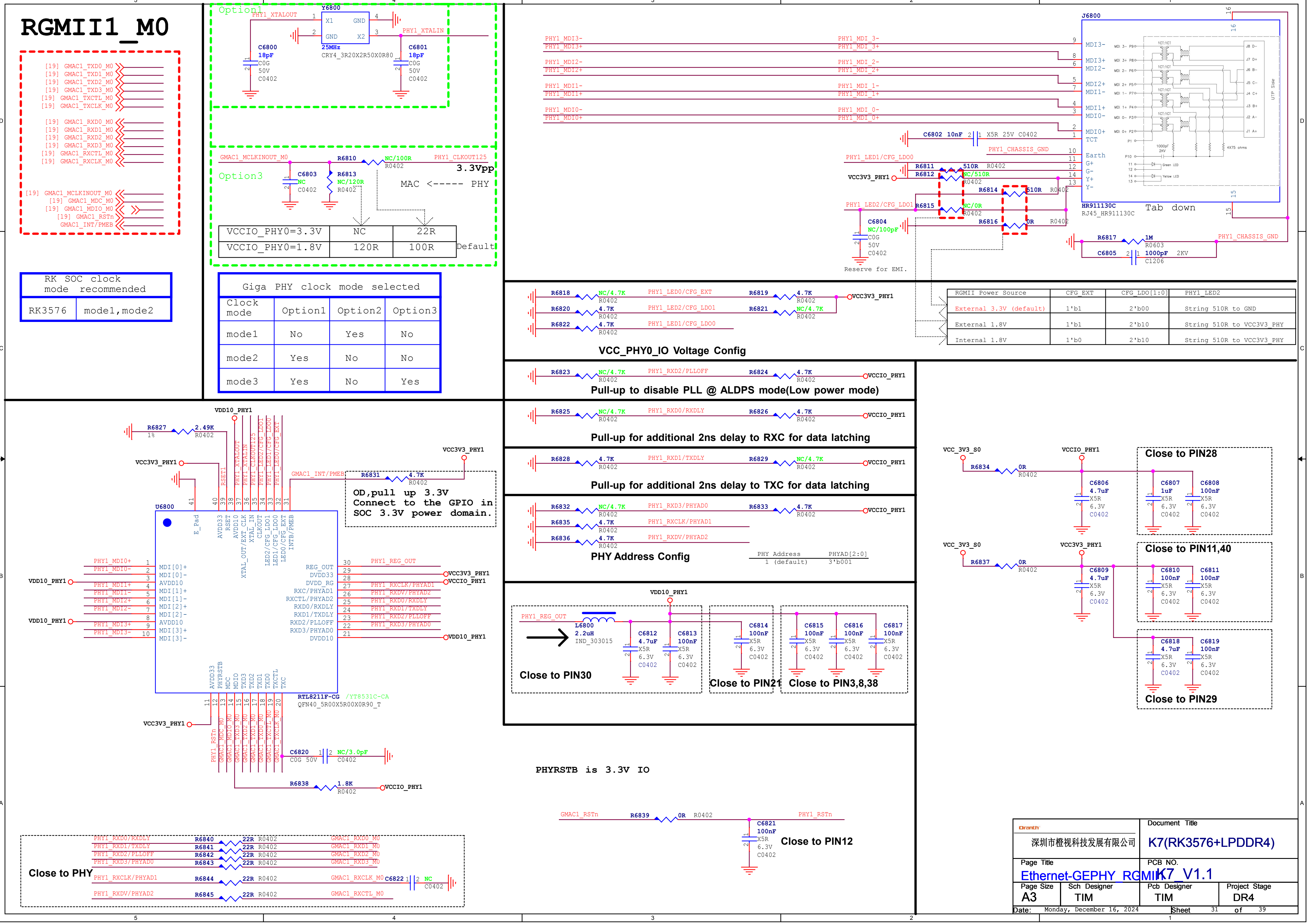

RK SOC clock mode recommended	
RK3576	mode1, mode2

VCCIO_PHY0		
VCCIO_PHY0=3.3V	NC	22R
VCCIO_PHY0=1.8V	120R	100R

Default

Giga PHY clock mode selected

Clock mode	Option1	Option2	Option3
mode1	No	Yes	No
mode2	Yes	No	No
mode3	Yes	No	Yes

RGMII Power Source	CFG_EXT	CFG_LDO[1:0]	PHY1_LED2
External 3.3V (default)	1'b1	2'b00	String 510R to GND
External 1.8V	1'b1	2'b10	String 510R to VCC3V3_PHY
Internal 1.8V	1'b0	2'b10	String 510R to VCC3V3_PHY

PHY Address	PHYAD[2:0]
1 (default)	3'b001

Orenth 深圳市橙视科技发展有限公司	Document Title: K7(RK3576+LPDDR4)
Page Title: Ethernet-GEPHY RGMII	PCB NO.: K7_V1.1
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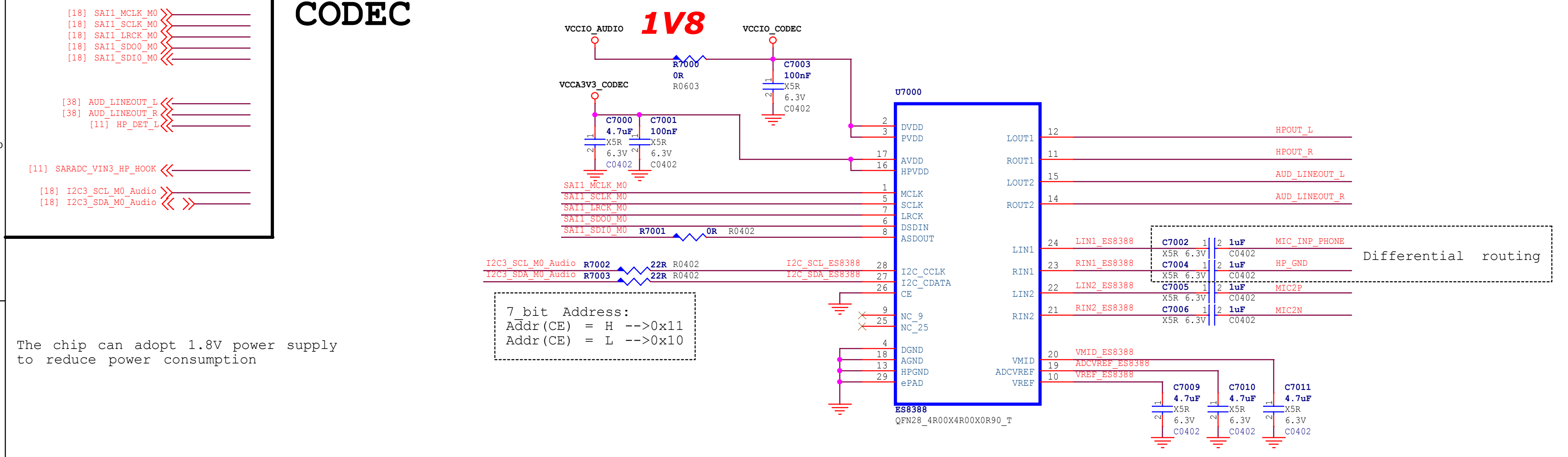
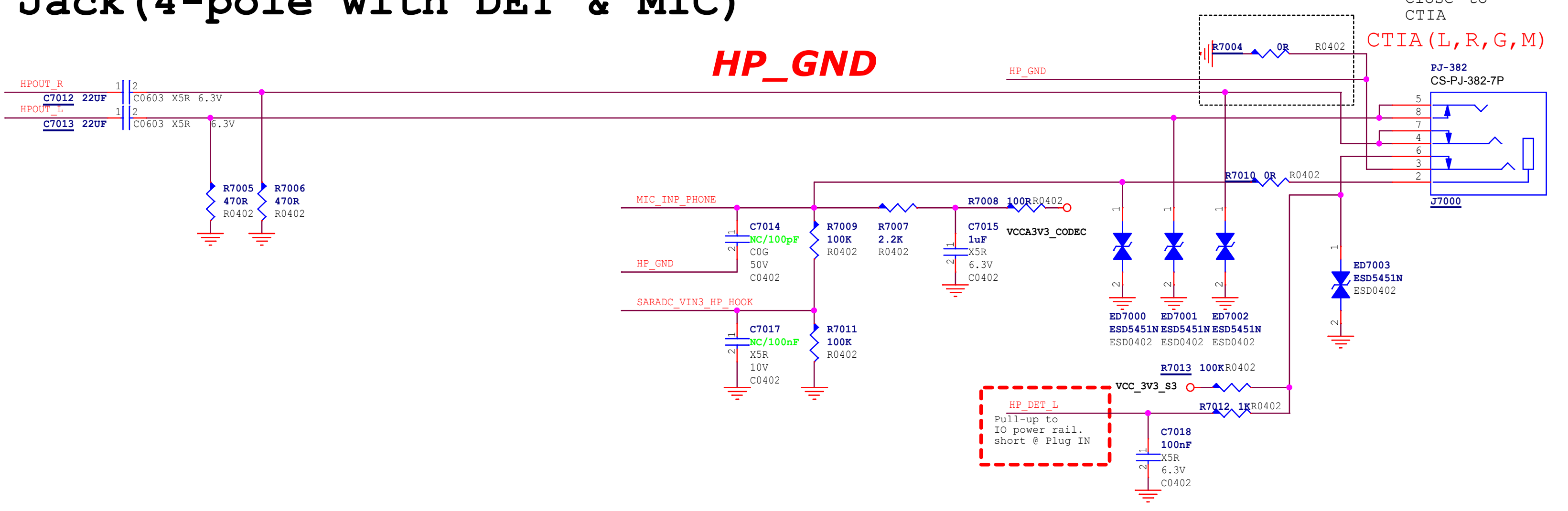
CODEC

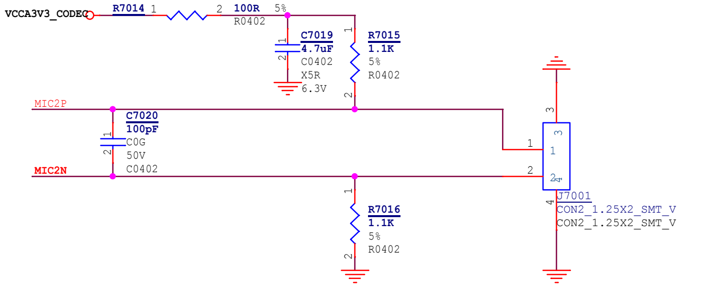


The chip can adopt 1.8V power supply to reduce power consumption

Headphone Jack(4-pole with DET & MIC)



MIC



深圳市橙视科技发展有限公司		Document Title: K7(RK3576+LPDDR4)	
Page Title: Audio-CODEC(ES8388)		PCB NO.: K7_V1.1	
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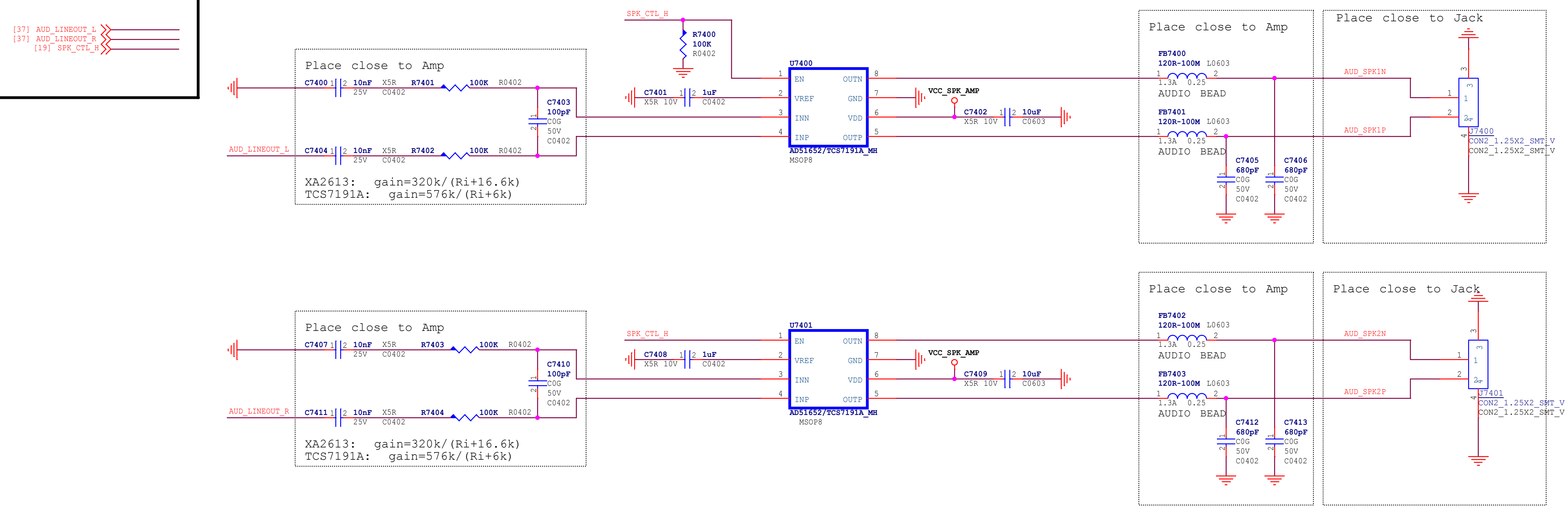
Speaker Output



深圳市橙视科技发展有限公司	Document Title: K7(RK3576+LPDDR4)
Page Title: Audio-SPK	PCB NO.: K7_V1.1
Page Size: A3 / Sch Designer: TIM	Pcb Designer: TIM / Project Stage: DR4
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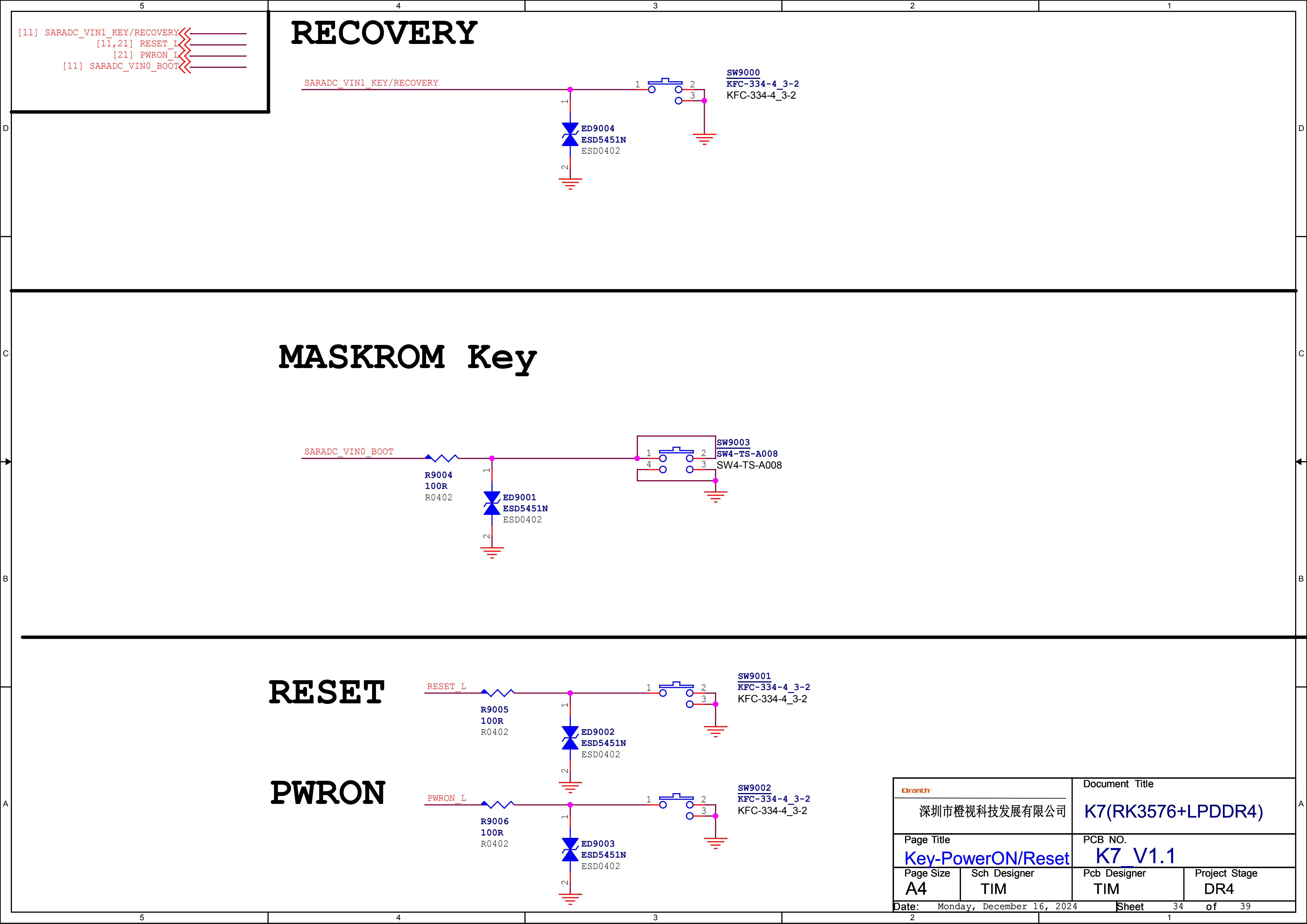

RECOVERY
[11] SARADC_VIN1_KEY/RECOVERY
[11,21] RESET_L
[21] PWRON_L
[11] SARADC_VIN0_BOOT
SARADC_VIN1_KEY/RECOVERY
SW9000
KFC-334-4_3-2
KFC-334-4_3-2
ED9004
ESD5451N
ESD0402
MASKROM Key
SARADC_VIN0_BOOT
R9004
100R
R0402
ED9001
ESD5451N
ESD0402
SW9003
SW4-TS-A008
SW4-TS-A008
RESET
RESET_L
R9005
100R
R0402
ED9002
ESD5451N
ESD0402
SW9001
KFC-334-4_3-2
KFC-334-4_3-2
PWRON
PWRON_L
R9006
100R
R0402
ED9003
ESD5451N
ESD0402
SW9002
KFC-334-4_3-2
KFC-334-4_3-2
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Document Title
K7(RK3576+LPDDR4)
Page Title
Key-PowerON/Reset
PCB NO.
K7_V1.1
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A4
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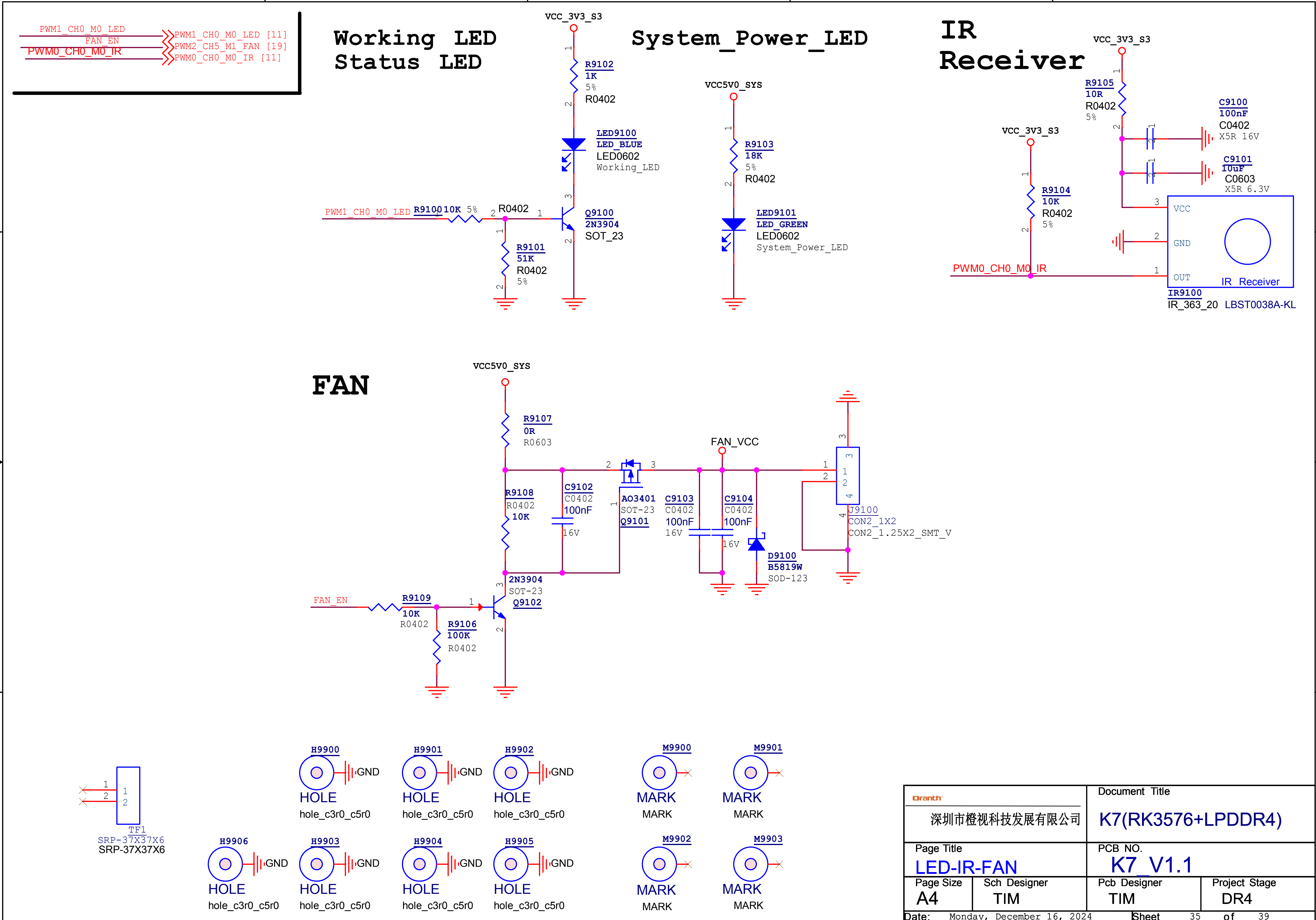

Oranth 深圳市橙视科技发展有限公司	Document Title: K7(RK3576+LPDDR4)
Page Title: LED-IR-FAN	PCB NO.: K7_V1.1
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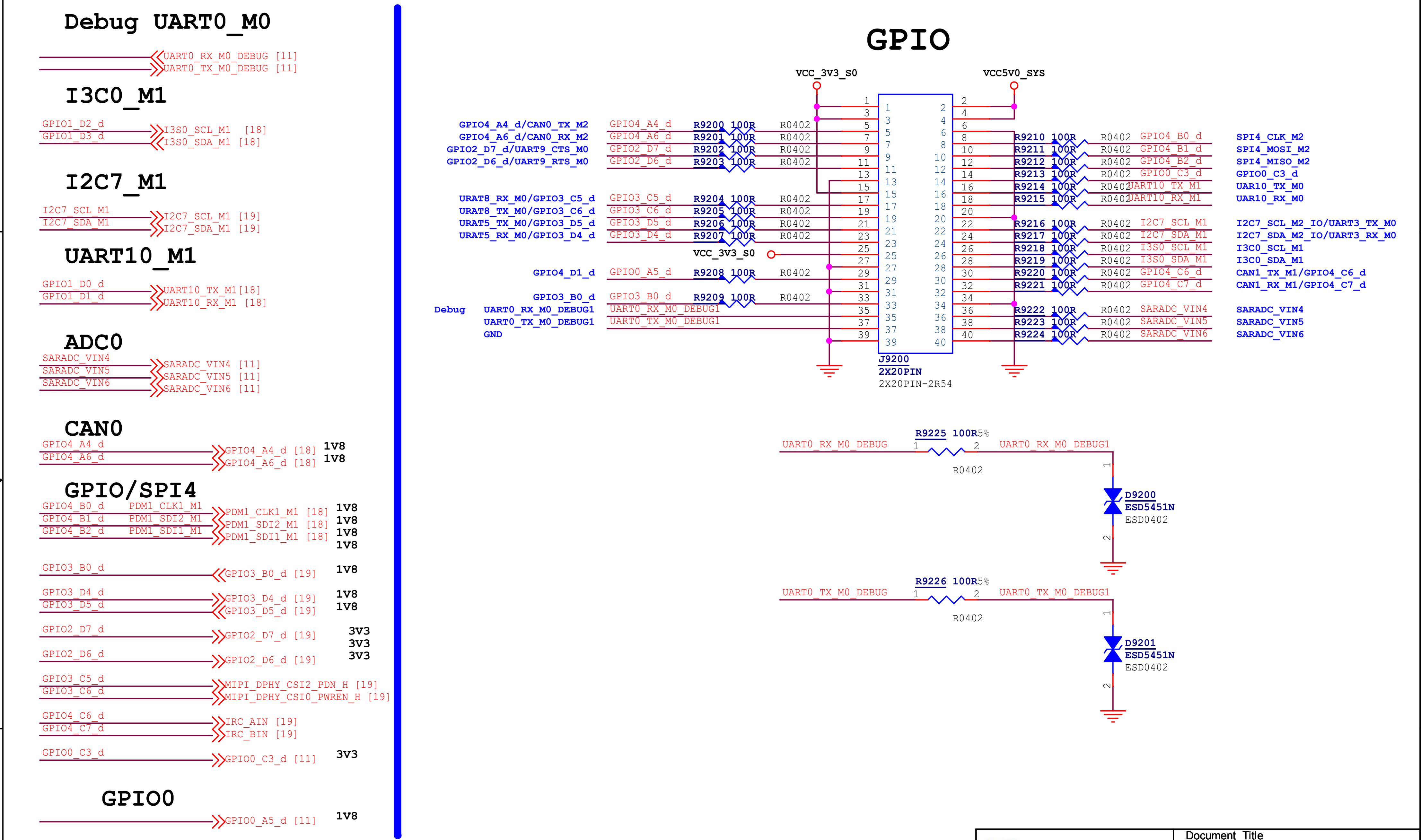

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Document Title	K7(RK3576+LPDDR4)		
Page Title	GPIO/Debug	PCB NO.	K7_V1.1
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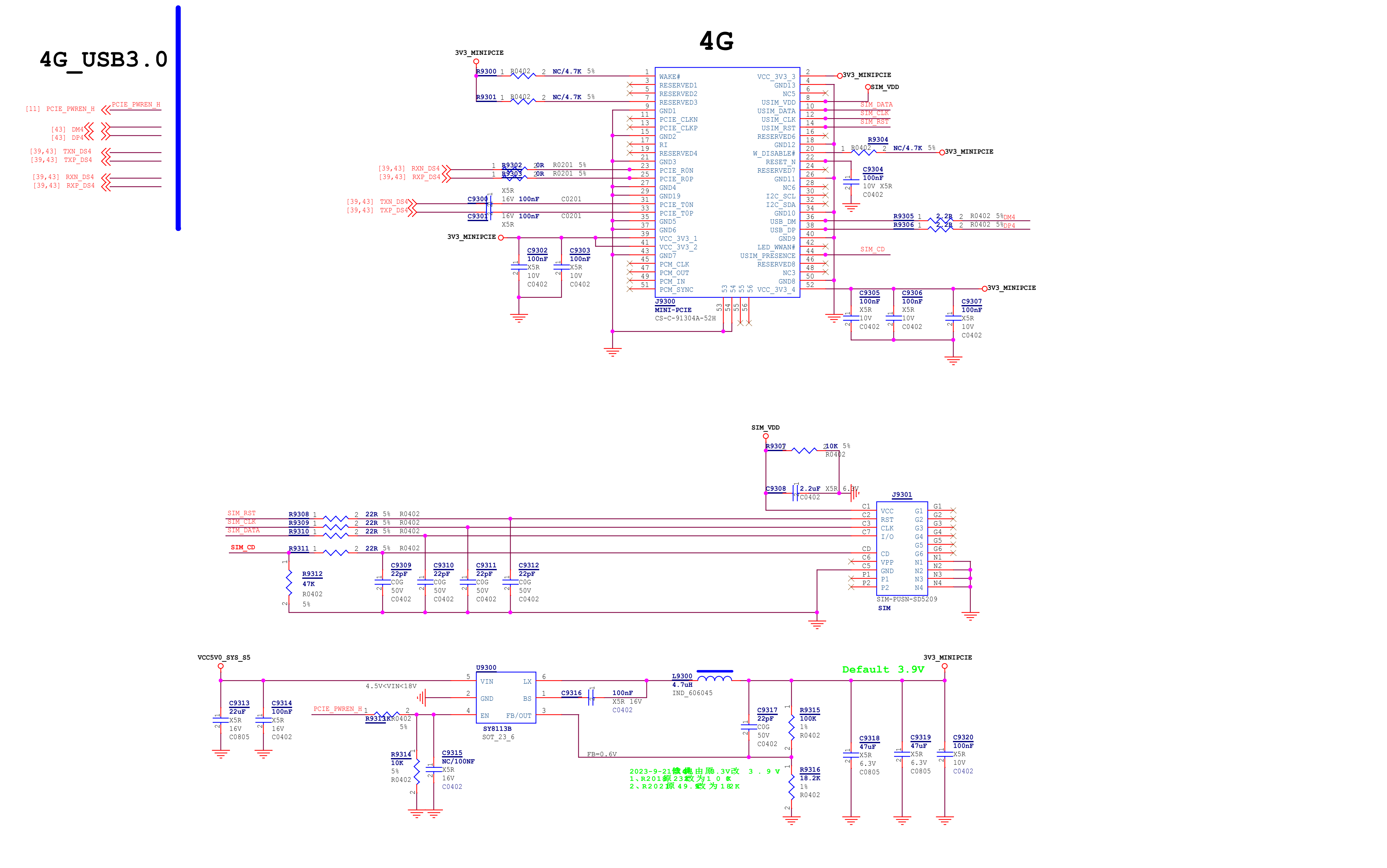

Orangeth 深圳市橙视科技发展有限公司		Document Title: K7(RK3576+LPDDR4)	
Page Title: 4G-USB30		PCB NO.: K7_V1.1	
Page Size: A3	Sch Designer: TIM	Pcb Designer: TIM	Project Stage: DR4
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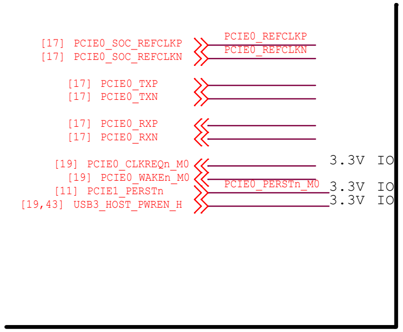
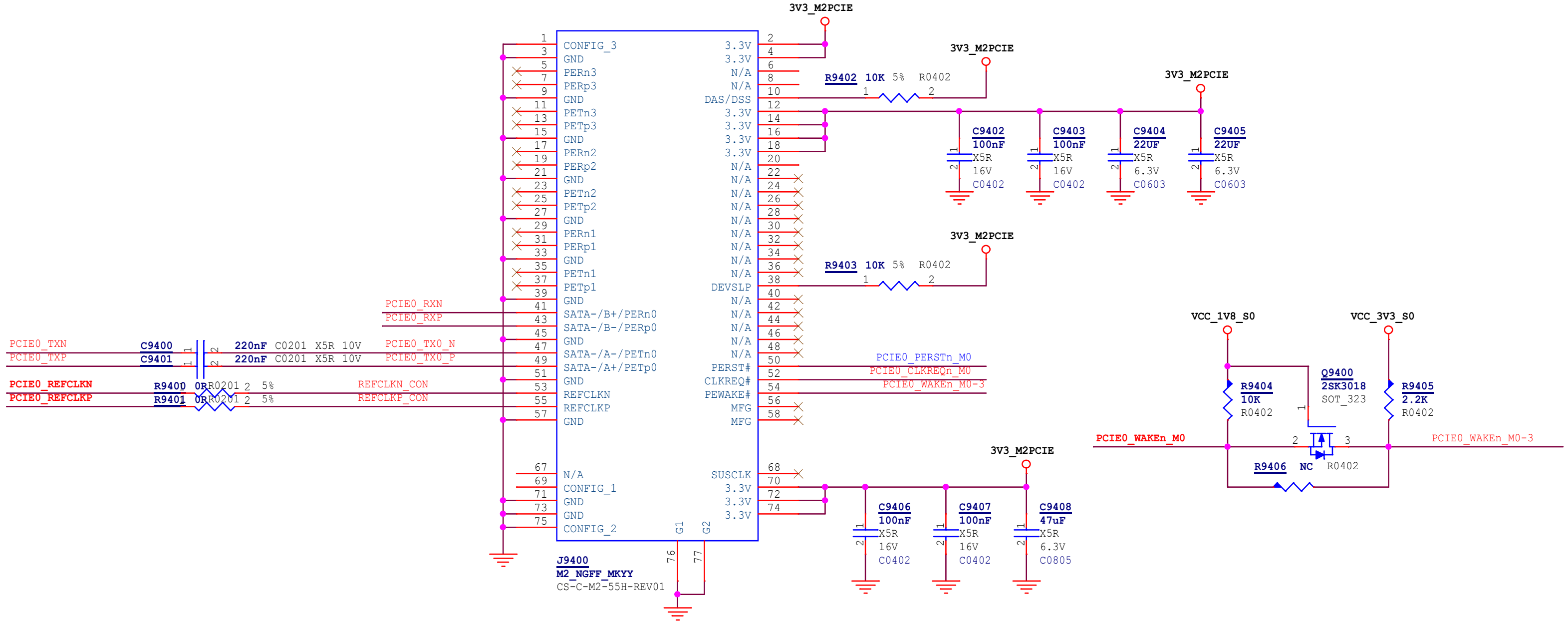
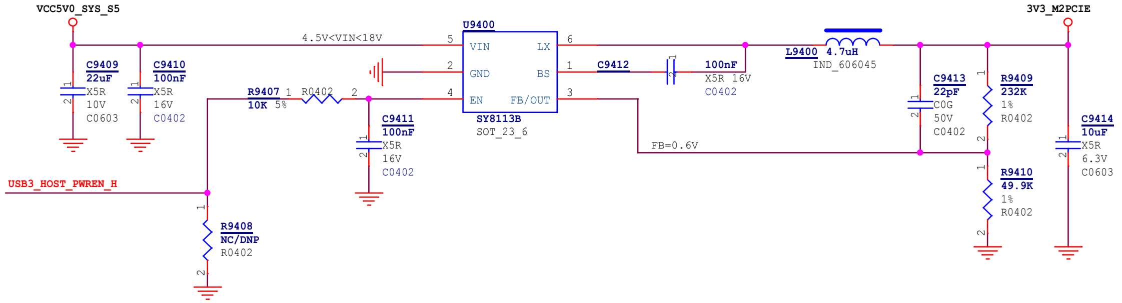


M.2 PCIe2.0 x 1Lanes



M2 POWER



深圳市橙视科技发展有限公司		Document Title: K7(RK3576+LPDDR4)	
Page Title: PCIe0_M2		PCB NO.: K7_V1.1	
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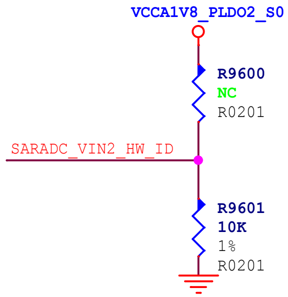
[11] SARADC_VIN2_HW_ID

HW_ID



Config Table for SARADC_VIN2_HW_ID

Item	Rup	Rdown	ADC Value	VERSION
HW_ID1	NC	10K	0	RK3576 EVB1 V12
HW_ID2	10K	1.13K	416	RESERVE
HW_ID3	10K	2.49K	816	RESERVE
HW_ID4	10K	4.3K	1231	RESERVE
HW_ID5	10K	6.8K	1658	RESERVE
HW_ID6	10K	10K	2048	RESERVE
HW_ID7	10K	14.7K	2437	RESERVE
HW_ID8	10K	23.2K	2862	RESERVE
HW_ID9	10K	40.2K	3279	RESERVE
HW_ID10	10K	88.7K	3680	RESERVE
HW_ID11	10K	NC	4095	RESERVE

SARADC PIN

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Pcb Designer: TIM	Project Stage: DR4
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